How computational complexity can restore general equilibrium in markets with indivisible goods*

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Abstract

Walrasian Equilibrium (WE) may fail to exist when goods are indivisible. We propose the Complexity Compensating Equilibrium (CCE), in which prices render consumers' budget problem instances computationally most difficult, so demands can become heterogeneous because agents apply different cognitive effort or are endowed with varying cognitive capability. We show that CCE can exist even where WE does not, and we derive and test its necessary conditions. In a controlled experiment, trade prices are high despite zero marginal cost on the supply side, and tend to locate consumers' budget problem instance in regions of high computational difficulty. We further find that prices are highly volatile, forcing agents to repeatedly solve a relatively small NP-hard problem under severe time constraints. When WE exists, observed prices deviate significantly from it and final holdings imply markedly different earnings levels, rejecting WE in favor of CCE.

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1 Introduction

A long-standing question in general equilibrium theory is whether and how markets equilibrate in the presence of indivisible goods. With homogeneous preferences and symmetric and tight individual budgets, Walrasian Equilibrium (WE) generically does not exist (Henry, 1970).To illustrate why, consider the simplest case of two agents and one unit of an indivisible good. Assume that both agents value the good at ϕ and have budgets of C with $\phi > C$. For any price $p \leq C$, both agents demand the good, and for any price $p > C$, neither agent can afford the good. Thus, there is no price that clears the market.

Existence of general equilibrium in pure exchange economies involves three conditions: (i) budget feasibility (allocations satisfy budget constraints), (ii) optimality (utility is maximized given preferences and budgets), and (iii) market clearing (demand to equal supply in all markets). When indivisibilities stand in the way of general equilibrium, attempts to restore existence relax one of the standard assumptions. For instance, the economics literature has shown that general equilibrium can be restored if preferences exhibit a sufficient degree of substitutability or complementarity, if individual budgets are sufficiently heterogeneous, or if market clearing is allowed to be approximate. Rather than restoring equilibrium by relaxing preference or budget homogeneity, we instead relax the assumption that agents perfectly solve their optimization problem, in the spirit of a smaller literature that proposes alternative equilibrium concepts (see Appendix H for a detailed literature review).

This article makes three contributions. First, we propose a new equilibrium concept, the Complexity Compensating Equilibrium (CCE), which acknowledges the NP-hardness of the budget problem when goods are indivisible, and hence, the need to spend substantial cognitive effort or to be endowed with better cognitive capability to find the utility maximizing choice. Second, we provide experimental evidence supporting the predictions generated by CCE. Third, we pit our equilibrium against WE when the latter exists and provide experimental evidence in favor of CCE and against WE.

CCE builds on the intuition that when goods are indivisible, agents face a computationally complex decision problem, namely, an NP-hard problem (Gilboa et al., 2021).[1] In a typical competitive market, the agent knows the budget as well as the payoffs (personal values) and prices of the goods. Even though this information is minimal, deciding which bundle is optimal is far from trivial. Our conjecture is that when the complexity of agents' individual budget problem instances is high, agents with homogeneous preferences differ widely in their willingness to spend cognitive effort or their cognitive capability.[2] As a result, they will select different consumption bundles, associated with different utility levels. This stratification allows markets to equilibrate.

To distinguish between instances of the budget problem with high and low complexity, we exploit the existence of a *phase transition* in the *decision version* of the budget problem. In the decision version, the goal is to determine whether utility can be improved by switching to another feasible choice. If so, the instance is said to be *solvable*. The phase transition is the region where utility is increased so that the decision instances of the budget problem change from almost always solvable to almost never solvable. In the phase transition, the number of computations needed to solve an instance is high on average, for both electronic computers and humans.[3] While NP-hardness is formally an asymptotic notion, empirical research on humans in combinatorial optimization tasks shows that, near phase the phase transition, accuracy in finding optimal solutions drops even in small instances (Yadav et al., 2020; Franco et al., 2021). In a realistic markets context, this difficulty is bound to be exacerbated because prices change, and hence, the agent may need to re-compute desired holdings periodically. We argue therefore that, to equilibrate, prices will locate the optimal budget allocation in the phase transition. This way, only those who apply sufficient cognitive effort or are endowed with sufficient cognitive capability will find the optimum, while others will have to settle for second-best (or worse) allocations.

We also rely on a second measure of instance complexity in NP-hard problems,

[1]NP is an acronym that stands for "Nondeterministic Polynomial." Explanation of the origin of this term would require us to get into Turing machines. We believe that this is not necessary for the reader to understand the gist of our arguments, so we refrain from elaborating. The interested reader can consult standard textbooks, such as Arora and Barak (2009), or, perhaps more pertinent for economists, Bossaerts (2026).

[2]It is useful to view an agent's performance on the budget problem as the output of a production function whose two inputs are the agent's cognitive capability—a productivity endowment—and the cognitive effort the agent applies. As with any such production function, a given level of output can be reached by trading off one input against the other, so an agent can attain the optimum through sufficient capability, sufficient effort, or a combination of the two.

[3]Phase transitions exist for other NP-hard problems. Among others, Yadav et al. (2020) demonstrate existence of a phase transition in the 0-1 Knapsack Problem (KP), closely related to the budget problem. In the budget problem, cash can be used to exhaust one's budget, while in the KP, there is no cash; all items are indivisible. The relation is made precise in Section 3.

which is the count of optimal solutions, technically referred to as the number of *witnesses*. Because of heterogeneity in search paths, humans are known to perform better when multiple witnesses exist (Franco et al., 2021). To ensure maximum complexity, we therefore expect markets to select price configurations so that the budget problem has a minimal number of solutions.

Together, these two measures characterize the region in which the budget problem instances are computationally hardest. CCE posits that prices will push the budget problem instances into this region, so that only those who spend sufficient cognitive effort or are endowed with sufficient cognitive capability will find the optimum. Demands become heterogeneous, thus enabling equilibration.

On top of these computational requirements for existence of CCE, we impose traditional market clearing conditions that are necessary given the scarcity of available goods. If, as will be the case in the experiment we run to verify our theory, there are only two units of three indivisible goods per three agents, then all goods pairs will have to be affordable, but nobody should be able to afford all three goods.

Unique to CCE, however, is the necessary condition that different levels of utility accrue to the various bundles that agents choose to hold. There cannot be more than one bundle that reaches maximal utility unless equal cognitive effort or capability to compute another optimum is required. Other bundles that are held in equilibrium must earn less utility because they require less cognitive effort or capability. This will be in sharp contrast with the predictions of WE (when it exists) as we shall point out later.

We limit our attention to verifying the necessary conditions of equilibrium. A full equilibrium characterization would require a detailed model of cognitive effort allocation or a specification of the distribution of cognitive capabilities across agents and a production function that allows the agent to trade off effort against capability. At present, relatively little is known about cognitive effort in such problems or cognitive capability, in part because humans appear to resort to a wide range of algorithms (heuristics), both across individuals and over time. Empirical evidence from the closely related 0-1 Knapsack Problem shows substantial heterogeneity in search behavior and limited exploration of the feasible choice set (Meloso et al., 2009; Murawski and Bossaerts, 2016; Bossaerts et al., 2024).[4] Moreover,

[4]In KP with 10 goods, Murawski and Bossaerts (2016) report that, individually, 20 participants visit only 3.6% of all feasible knapsacks, while collectively visiting 42.1%. The heterogeneity in search approaches suggests that communication should be beneficial. This has proven to be correct; purposely designed markets manage to substantially improve individual performance, thus spreading knowledge in society, as originally predicted in Hayek (1945). See Meloso et al.

effort allocation at times appears to defy economic intuition: as the probability to find the optimum decreases, the utility attained nevertheless increases (Murawski and Bossaerts, 2016; Andrabi et al., 2026). Our objective instead is to characterize the economic and computational conditions under which CCE achieves market clearing while remaining compatible with homogeneous preferences and budgets. Accordingly, our empirical object is not a unique equilibrium price vector but a region of price space characterized by necessary computational and economic restrictions.

Nevertheless, to illustrate the structure of CCE equilibria, and to show that such an equilibrium *can* exist when WE does not, we introduce an example (see Section 2) where we assume that agents choose an algorithm from a family of algorithms referred to as the *Sahni-k family.* The family has been used to construct a metric of instance complexity that correlates with human performance (Meloso et al., 2009; Murawski and Bossaerts, 2016; Andrabi et al., 2026), the performance of nonhuman primates (Hong and Stauffer, 2023), and the informational efficiency of prices in financial markets (Meloso et al., 2009; Bossaerts et al., 2024).[5] Although Sahni-k family provides a good framework to elucidate how humans deal with the 0-1 Knapsack Problem (and hence, the budget problem, which is only a special case), it is far from all-encompassing; see Andrabi et al. (2026) for discussion.

It is insightful to consider the case where WE *does* exist (despite indivisibilities) and compare its predictions with those from CCE, both in terms of final holdings and in terms of prices. Under indivisibilities existence of WE typically requires there to be multiple optimal goods bundles, *all delivering the same level of utility.* The opposite happens in CCE: agents choose bundles with differing utility; which bundle they end up choosing and hence which utility they end up with depends on how much cognitive effort they are willing or able to apply. Thus, under CCE, we expect distinct categories of participants separated by the utility of their final holdings. The equilibrium budget problem instance in WE may reside inside the phase transition. But in our experiment, we ensured that it located away from

(2009) and Bossaerts et al. (2024) for details.

[5] An alternative approach would be to define the competitive equilibrium as obtaining under *noisy optimization* – in analogy with a popular equilibrium modeling technique in game theory; see, e.g., Mckelvey and Palfrey (1998). However, we do not consider the deviations from optimality to be purely random. Indeed, in the context of computational complexity, human effort and performance (relative to the optimum) exhibit structure that can be analyzed using the tools of the theory of computational complexity; see citations in main text. This structure enables better modeling than what can be obtained by merely assuming that the mistakes – relative to optimum – are random.

(below) the phase transition in order to increase the power to distinguish CCE and WE in the data.

To provide empirical evidence for CCE, we conduct a controlled laboratory market experiment with three indivisible goods and cash. As is standard in experimental research on general equilibrium theory (e.g., Bossaerts et al. (2007); Asparouhova et al. (2016)), the market institution we use is a continuous double-sided open-book in which both consumers and sellers submit limit orders that remain unless they are cleared (traded) or they are canceled. Orders are cleared using price-time priority matching. The aggregate supply and demand are chosen so that there are two units of a good available per three consumers. Consumers (buyers/the demand side) are endowed with and value cash but can receive pre-announced payoffs for the first unit of each good they acquire in the marketplace, thereby increasing their take-home earnings if they can acquire it at a price below payoff. The sellers (supply side) initially hold the entire supply of the goods but receive no payoff for them; only cash has value for them, so they want to sell as much as possible, at any price. Their marginal cost is zero.

We implement two treatment variations. First, to ensure that our findings are robust to changes in the numerical values of the budget problem we vary within participants the goods payoffs. Each session consists of 16 trading periods split into two blocks of eight periods, and the payoff configuration is fixed within each block. The order of the two blocks is counterbalanced between sessions. Second, we vary across participants, the cash endowment (income/budget) of the consumers. In the low-income treatment WE does not exist, while it does in the high-income treatment. Between 16 and 20 participants participate in each session. We thus present experimental evidence from 180 participants and 2,880 participant-periods. In total, participants submitted 17,628 limit orders resulting in 4,121 trades.

Whether in CCE or WE, in our setting, equilibrium requires that all goods are transferred from the sellers to the consumers since the former face zero marginal cost while the latter have positive utility for the first unit of a good. Equilibrium also requires that prices are sufficiently high so nobody can afford all goods, implying that sellers will earn substantial surplus. But past experiments with a single good (plus cash) and a supply side with zero or low marginal costs have invariably produced prices that were biased towards zero, far below equilibrium predictions (Smith and Williams, 1982; Rasooly, 2022). If the findings in these simple partial-equilibrium experiments have any bearing on our general-equilibrium multi-market setting, then our design effectively stacks the deck against finding any evidence of equilibrium, whether CCE or WE.

We find overwhelming support for both the computational and economic requirements of CCE. When WE exists, it does not emerge: trade prices and utilities attained continue to be consistent with CCE. When prices move away from the region where they satisfy necessary computational and/or economic conditions for CCE, vector autoregression (VAR) analysis shows that they tend to revert. When the values (utility levels) of the equilibrium goods bundles are too close, VAR analysis demonstrates that they tend to diverge again, consistent with CCE. Importantly, our data show that sellers manage to extract massive economic rents, despite incurring zero marginal cost, and in contrast to the aforementioned evidence from partial-equilibrium experiments.

At the same time, we discover huge volatility in trade prices. This is in contrast with observation from simple experiments with one, divisible good (and cash) (Smith, 1962; Smith and Williams, 1982). The volatility exacerbates the complexity of the budget problem that the demand side faces: by the time a consumer has figured out an optimal allocation, the opportunities to trade at prices she used to compute the optimum may have disappeared and she has to return to computing an optimum for the new price configuration, wait (and hope) that the original opportunities return, or settle for second-best (or worse).

While a budget problem with only four goods (including cash) may sound easy,[6] in practice the speed with which prices change makes choices very challenging. Consumers are asked to either solve small instances of the budget problem under immense time constraint, or posit prices, solve the resulting instance, and then engage with the market to obtain the solution at the posited prices. It thus appears that markets use price volatility to further exploit complexity as a way to equilibrate in the face of indivisibilities. In future work, we plan to increase the number of goods traded. We conjecture that price volatility would decline: markets no longer need to put time constraints on solving the budget problem since the budget problem instances will have become more difficult.

The remainder of the paper is organized as follows (a detailed discussion of related literature on equilibrium existence under indivisibilities is provided in Appendix H). Section 2 provides a numerical example of a CCE based on stylized assumptions of effort allocation and ability, while Section 3 presents necessary computational and economic requirements for CCE that do not require us to un-

[6]We will point out later, however, that one can readily generate complex instances with only four goods that are as difficult—for humans, that is—as simple instances with eight goods. We will introduce the necessary measures of difficulty to back up such a claim.

derstand how exactly humans solve the budget problem. Section 4 discusses our experimental design in full detail and identifies testable predictions for the necessary conditions of CCE. Results are analyzed in Section 5, and we provide a concluding discussion in Section 6.

2 An illustrative numerical example

Consider an economy with identical agents, with additive utility in goods and cash, and unit demand for goods. The agents compete to obtain *three* possible goods: (i) a holiday to visit family in Spain, with a payoff of 1,200, (ii) an electric bike, with a payoff of 2,400,[7] and (iii) a professional camera kit, with a payoff of 3,000.[8] The payoffs can be interpreted as the monetary equivalent of the consumption utility of each good. Agents only value the first unit of a good they acquire. We denote the three goods as L, M, and H indicating low, medium, and high payoff respectively. The supply for each good is equal to *two* for every three agents, and they are indivisible. A market equilibrium in this economy would require one third of the agents to purchase goods $\{M, L\}$, one third to purchase goods $\{H, L\}$, and one third to purchase goods $\{H, M\}$.[9]

Traditional WE prices would be such that the agents are indifferent between all possible pairs of goods. The minimum budget for WE to exist is 3,000.[10] If we assume that the agents have 3,200 in cash and a tick size of 50,[11] then the unique

[7]Note that the indivisibility of the bicycle can be resolved by introducing e-bike renting. Interestingly, however, the introduction of e-bike renting in reality has not solved the problem of indivisibilities, because rentals generally require subscriptions. The subscription contract reintroduces indivisibility.

[8]Like with the bicycle, rental contracts could also be introduced to overcome indivisibility in the case of the camera kit. There too, a subscription may be required however. But it is impossible to imagine a way to make the visit to family in Spain perfectly divisible. One cannot opt for a halfway trip, since that would defeat the purpose of the visit.

[9]To see this, remember that consumers are identical, and hence face the same budget constraint. If some consumers obtain a basket with only one good (say, H), and others baskets with two or three goods, H must be priced such that the former group cannot afford a second good. But since all agents have the same budget, the price of H must also be too high for the latter group. Equilibrium does not obtain because only one unit of H is demanded per three agents, while two units of H are supplied per three agents. Similarly, if some consumers buy nothing and the others all three goods, all the goods must be sufficiently expensive for the former group not to buy anything, meaning no one can afford the full basket. Therefore, for total demand to equal total supply, the number of agents holding each of the possible pairs of goods must be equal.

[10]Appendix A provides a formal derivation of the minimum budget for Walrasian equilibrium existence, and restrictions on the price range for each good when equilibrium is not unique.

[11]The *tick size* is the distance between allowable prices, usually starting from a single tick. Tick sizes are universally imposed in organized financial markets.

equilibrium prices would be 100 for the holiday (L), 1,300 for the bike (M) and 1,900 for the camera (H). For those prices, the agents obtain a profit (surplus) of 1,100 from each good and receive a total utility of 5,400 from the equilibrium pairs.

What has not been widely appreciated in the literature yet is that, with indivisibilities, the budget problem is difficult; in the language of computer science and complexity theory it is NP-hard. (We will prove this in the Appendix B; Gilboa et al. (2021) prove this for a related problem, which they refer to as the consumer problem.) Our equilibrium concept, CCE, takes this into account. We propose that agents use different algorithms in their attempt to solve their budget problems. These require different levels of cognitive effort and capability. The more effort required, the closer an algorithm gets to the optimum. One can think of one of these algorithms as a *heuristic*. Some heuristics are better than others.

A family of approximation algorithms that has been found to explain to some degree how humans choose in the KP (0-1 Knapsack Problem) is the Sahni-k class, derived from the Sahni-Horowitz algorithm (Sahni, 1975; Murawski and Bossaerts, 2016; Andrabi et al., 2026). The simplest ($k = 0$), lowest-effort member of this class is the greedy algorithm, whereby goods are sorted in descending order of the ratio value/price, defined as the *density*, and the goods are purchased until exhausting one's budget.[12]

For $k = 1$, the algorithm is altered by cycling through all feasible initial allocations of one item, followed by application of the greedy algorithm on the remaining items, retaining eventually only the choice that maximizes total value. For $k = 2$, the algorithm cycles through all feasible pairs of items, followed by application of the greedy algorithm, again retaining only the highest-value choice. The algorithm is defined analogously for higher values of k. Clearly, *cognitive effort increases in k*. But *performance also increases in k*. With $k = 0$ the algorithm can reach the optimum only if the instance can be solved correctly by the greedy algorithm. With $k = I - 1$, where I is the total number of goods, the optimum is always found.

The following statistic offers an idea of how k affects the ability of humans to solve the KP: when comparing two instances where the k required to reach the optimum is increased by one, value attained (as a percentage of the optimal value)

[12] More sophisticated versions exist, whereby, e.g., items are skipped when they violate the budget constraint as long as there are remaining items that do fit within the budget. It deserves emphasis that the greedy algorithm finds the optimal solution if all goods are infinitely divisible, but not if some goods are indivisible, or if only one good is infinitely divisible (as with cash remaining in one's budget in our example).

decreases as much as increasing the number of available items by *five* units. That is, with three items, going from $k = 2$ to $k = 3$ is equivalent to solving instances with eight items. See Bossaerts (2026).

For our example, we posit that agents sort into categories defined by k. Assuming agents have a budget of 3,200, this sorting could lead to the following CCE. Referring to the prices in Table 1a, the greedy algorithm would rank the holiday (L) first, followed by the bike (M), and the camera (H) would be ranked last. An agent using the greedy algorithm ($k = 0$) first selects L, and then proceeds with M. Their budget is insufficient for H, so they end up with the pair of $\{M, L\}$ goods. An agent using the algorithm with $k = 1$ preselects a good H which was not in the greedy solution, and then uses the greedy algorithm to select the second good. Given the ratios, that second good is L, so they end up with the pair of $\{H, L\}$ goods. Finally, an agent using the algorithm with $k = 2$ preselects H as the first good, and also preselects M as the second good. Their budget is insufficient for more goods, so they end up with the pair of $\{H, M\}$ goods. If one third of agents use the greedy algorithm ($k = 0$), one third uses the Sahni algorithm with $k = 1$, and one third uses the Sahni algorithm with $k = 2$, then the markets clear. A CCE is reached.

TABLE 1 Examples of CCE: (a) when WE exists, (b) and when WE does not exist

Market outcomes				Individual outcomes		
Good	Payoff	Price	Density	Sahni-k	Bundle	Utility
L	1,200	500	2.400	$k = 0$	(M,L)	5,000
M	2,400	1,300	1.846	$k = 1$	(H,L)	5,200
H	3,000	1,700	1.765	$k = 2$	(H,M)	5,600

(a) Budget 3,200

Market outcomes				Individual outcomes		
Good	Payoff	Price	Density	Sahni-k	Bundle	Utility
L	1,200	400	3.000	$k = 0$	(M,L)	4,800
M	2,400	1,200	2.000	$k = 1$	(H,L)	5,000
H	3,000	1,600	1.875	$k = 2$	(H,M)	5,400

(b) Budget 2,800

An important feature of our equilibrium, which also justifies the name, is that the agents who spend the highest effort find the optimal solution and gain the most (buying H and M); those who spend moderate effort find a good solution and gain less (buying H and L); and those who spend the least effort find the worst solution and gain the least (buying L and M). As can be seen from the last column of Table 1a, the agents are compensated for the cognitive effort of using a more sophisticated algorithm, or for their higher cognitive capability. The agent who uses $k = 2$ receives a utility of 5,600, the agent who uses $k = 1$ receives 5,200, and the agent who uses $k = 0$ receives 5,000. This is in sharp contrast with WE, which ignores the cognitive effort or capability needed to make better choices and requires prices to ensure indifference between all equilibrium demand bundles. Thus, in CCE, larger cognitive effort or capability is compensated for by higher utility. Notice also that agents applying maximal effort ($k = 2$) obtain a utility level that surpasses that of WE (5,600 *vs.* 5,400).

Now consider the case where the budget is 2,800, in which case WE does not exist. With tighter budgets, one would arguably expect prices for all goods to decrease. Consider such prices in Table 1b. The greedy algorithm still ranks good (L) first and good (H) last. If the agents use algorithms as before, then CCE continues to exist. We also note that while all pairs yield lower utility than before, there is still a cognitive premium so that the agents who demand the optimal bundle get the highest utility, of 5,400, and the agents who demand the second-best and third-best bundles receive lower utilities, of 5,000 and 4,800 respectively.

3 Complexity Compensating Equilibrium

We now introduce empirically testable necessary conditions for CCE. The central idea is that markets select a region of price space, rather than a unique price vector, that make budget problem instances sufficiently complex. At such prices, agents who differ in cognitive effort or capability choose different consumption bundles, with only those exerting sufficient effort or ability reaching the optimum. Cognitive effort or capability thereby induce demand heterogeneity.

Definition 1 (Complexity Compensating Equilibrium Region)**.** *Given I indivisible goods with payoffs $\phi = (\phi_1, \dots, \phi_I)$, a common budget C, aggregate supply S, and a feasible set of bundles, a* Complexity Compensating Equilibrium Region *is a set of price configurations*

$$\mathcal{P}^{CCE} \subseteq \mathbb{R}^I_+$$

such that, for prices $p \in \mathcal{P}^{CCE}$, there exists a distribution over selected bundles satisfying the following necessary conditions:

1. ***Computational difficulty.*** *The induced budget problem instance lies in a region of high computational difficulty, as measured by its location relative to the phase transition of the associated decision problem.*

2. ***Scarcity of optima.*** *The induced budget problem instance has few optimal solutions, with uniqueness as the generic case.*

3. ***Utility-ranked equilibrium bundles.*** *The bundles selected in equilibrium yield distinct utility levels, consistent with compensation for the cognitive effort or for cognitive capability required to identify higher-utility allocations.*

These conditions do not afford a full existence theorem since we do not specify cost of cognitive effort or endowment of cognitive capabilities. Rather, they characterize the region of price space in which CCE can obtain and generate empirically testable implications.

We first formalize the budget problem, discuss the computational complexity of solving instances of it, and show existence (and location) of a phase transition where the most difficult instances are expected to reside, and hence where we expect CCE to locate. We then use this structure to state the computational and economic requirements for CCE.

3.A The budget problem with indivisible goods

Consider an economy with I indivisible goods, each fully characterized by a *price* p_i and a *payoff* ϕ_i. There are N agents in the economy, each endowed with a budget (income) of C. Agents exhibit linear utility in goods and cash, and demand at most one unit of each good. Agents intend to choose the payoff maximizing bundle among all the affordable bundles. Since all goods are indivisible, an agent's choice can be simplified to a series of binary choice variables x_i, which take the value of 1 if good i is purchased and 0 otherwise.

Formally, each agent faces the following utility optimization problem.

$$\max_{\substack{x_i \in \{0,1\},\\ i=1,2,\ldots I}} \left\{ \sum_{i=1}^{I} x_i \phi_i + \left(C - \sum_{i=1}^{I} x_i p_i \right) \right\},$$

$$\text{subject to: } \sum_{i=1}^{I} x_i p_i \leq C.$$

The objective function can be rewritten as follows.

$$\max_{\substack{x_i \in \{0,1\},\\ i=1,2,\dots I}} \left\{ C + \sum_{i=1}^{I} x_i(\phi_i - p_i) \right\} = \max_{\substack{x_i \in \{0,1\},\\ i=1,2,\dots I}} \left\{ C + \sum_{i=1}^{I} x_i v_i \right\}. \tag{1}$$

The objective function is linear in the choice of goods, with *value coefficients* equal to $v_i = (\phi_i - p_i)$. We expect prices of chosen goods to be below goods payoffs ($p_i < \phi_i$), otherwise the agent prefers to hold cash. Therefore, the value coefficients are non-negative ($v_i \geq 0$). Written as in Equation 1, and ignoring the constant C, the budget problem is a special case of KP. Appendix B proves directly that the decision version of the budget problem is NP-hard.[13]

The hardness of the budget problem originates in the difficulty of verifying whether one has found the optimum. With infinitely divisible goods, and provided utility satisfies minimal smoothness conditions, determining whether one has reached an (interior) optimum merely requires to check first and second order conditions, i.e., to check how the solution changes in a small neighborhood. With indivisibilities, an NP-hard *decision* problem has to be solved instead, namely, whether there exists a solution with target value at least as large as the one already attained plus one (utility) unit. If so, the instance is said to be *solvable*, and optimum has not been reached yet.

We are interested in distinguishing difficulty among different *instances* of the budget problem. In the numerical example of the previous section, we assumed that agents use algorithms in the Sahni-k family to solve the budget problem. Fixing the algorithms agents use is a powerful tool to obtain concrete analytical results, but it is too strong of an assumption for empirical analysis for the reason cited before: we lack knowledge of how cognitive effort is allocated in budget problems, whether this allocation is done optimally, or how cognitive capabilities are distributed in the population. In order to derive testable implications, we instead resort to generic complexity measures of instances of the budget problem which do not depend on the algorithms used by the agents.

For KP, an *ex ante* metric of difficulty is whether the decision version of the instance is in the phase transition or not. With *ex ante*, we mean that it depends on features of the instance that can be recognized without any attempt at solving

[13] Gilboa et al. (2021) provides proof of NP-hardness of the consumer (decision) problem. Their problem is different from ours, since utility is expressed as a function of goods attributes, and these attributes map into goods. Also, the decision-maker does not receive utility from remaining cash in the budget.

it. With *phase transition*, we mean that there is a region where instances change from always-solvable to never-solvable.[14] Existence of such a phase transition for KP was first shown in Yadav et al. (2020).[15] There, it was also shown that the instances requiring most effort – for humans and electronic computers alike – reside in the phase transition. See also Franco et al. (2021, 2024); Andrabi et al. (2026).

While the budget problem can be re-written as a KP [see Equation 1], it is not obvious whether such a phase transition exists for the subset of budget problems we are interested in, where we fix item (goods) payoffs (ϕ_i in our notation). The reason why is that payoffs are agents' utilities for the individual goods, which the market cannot change; to alter difficulty, the market can only change prices, p_i.[16] In terms of the classical KP, this means that value coefficients v_i and costs p_i are negatively correlated since $v_i = \phi_i - p_i$. As such, we must first investigate whether KP instances where values and costs are negatively correlated also feature a phase transition.

The phase transition of the *budget decision problem* is defined as a region in a two-dimensional space. The dimensions of this space depend on the payoffs and the prices of the available goods, the budget, and a target utility Υ. The first dimension is the *normalized capacity* (κ), defined as the ratio of the budget to the total price of all goods. Informally, normalized capacity measures budget feasibility. Formally, it is defined as follows:

$$\kappa = \frac{C}{\sum_i p_i}. \tag{2}$$

The second dimension is the *normalized profit* (π), defined as the ratio of the target value Υ and the total value from purchasing all goods. Informally, normalized profit captures how difficult it is to find an allocation that reaches the target value when disregarding the budget constraint. As such, normalized profit captures

[14]An example of a phase transition in nature is the dew point: a small temperature interval where evaporation and condensation offset each other; for a higher temperature, evaporation dominates; at lower temperatures, condensation dominates.

[15]Many other computationally hard problems exhibit phase transitions, such as the satisfiability (3SAT) problem (Mitchell et al., 1992), the graph coloring problem (Cheeseman, 1991), the number partition problem (Gent and Walsh, 1998), and the traveling salesman problem (Gent and Walsh, 1996).

[16]Fixing payoffs may render the budget problem less difficult, moving it outside the set of NP-hard problems. This does not make NP-hardness less relevant for the agents, who may face different payoffs in different settings, and hence, who may need to choose an appropriate algorithm (a way to solve budget problem instances) that takes into account all possible payoff structures they may encounter. But here we analyze the budget problem from the point of view of the market, who faces a particular payoff structure and needs to select prices that ensure maximal difficulty.

value infeasibility. Formally, it is defined as follows:

$$\pi = \frac{\Upsilon}{C + \sum_i (\phi_i - p_i)}. \tag{3}$$

In the decision version of our budget problem, we ask whether a target value Υ can be reached or exceeded. If yes, then the instance is solvable; otherwise it is not. When the instance is in the phase transition, i.e., the target value, payoffs and prices are such that κ and π are in the phase transition, then uncertainty about solvability is highest and effort required to determine solvability tends to be highest.

Figure 1a displays, for each combination of κ and π, the fraction of decision problem instances that are solvable. Budget and payoffs are fixed as in one of the treatments in our experiment.[17] Shown are all instances for which prices are "*rational*," in the sense that prices are below payoffs, and prices are ranked in the same order as payoffs. Prices are forced to be on a finite grid, with tick size equal to 0.05. The tick size reflects the reality in financial markets—and in our experiment—that trade can happen only at discrete price levels.[18] For each price configuration, we test every feasible target.[19] We cut κ off at 1 (beyond which $\sum_i p_i > C$) for two reasons: (i) above $\kappa = 1$, the budget problem is trivial since the budget constraint is not binding, (ii) for equilibrium to obtain it is necessary that prices are sufficiently high so purchasing all items is infeasible.[20] We also cut π at 1.

Figure 1a reveals a distinct phase transition. Fixing κ while increasing the target value Υ – thereby increasing π and moving northwards in the plot – instances change from all-solvable (dark green) to non-solvable (dark red), with a small region where uncertainty about solvability is non-trivial (between light green and light red).

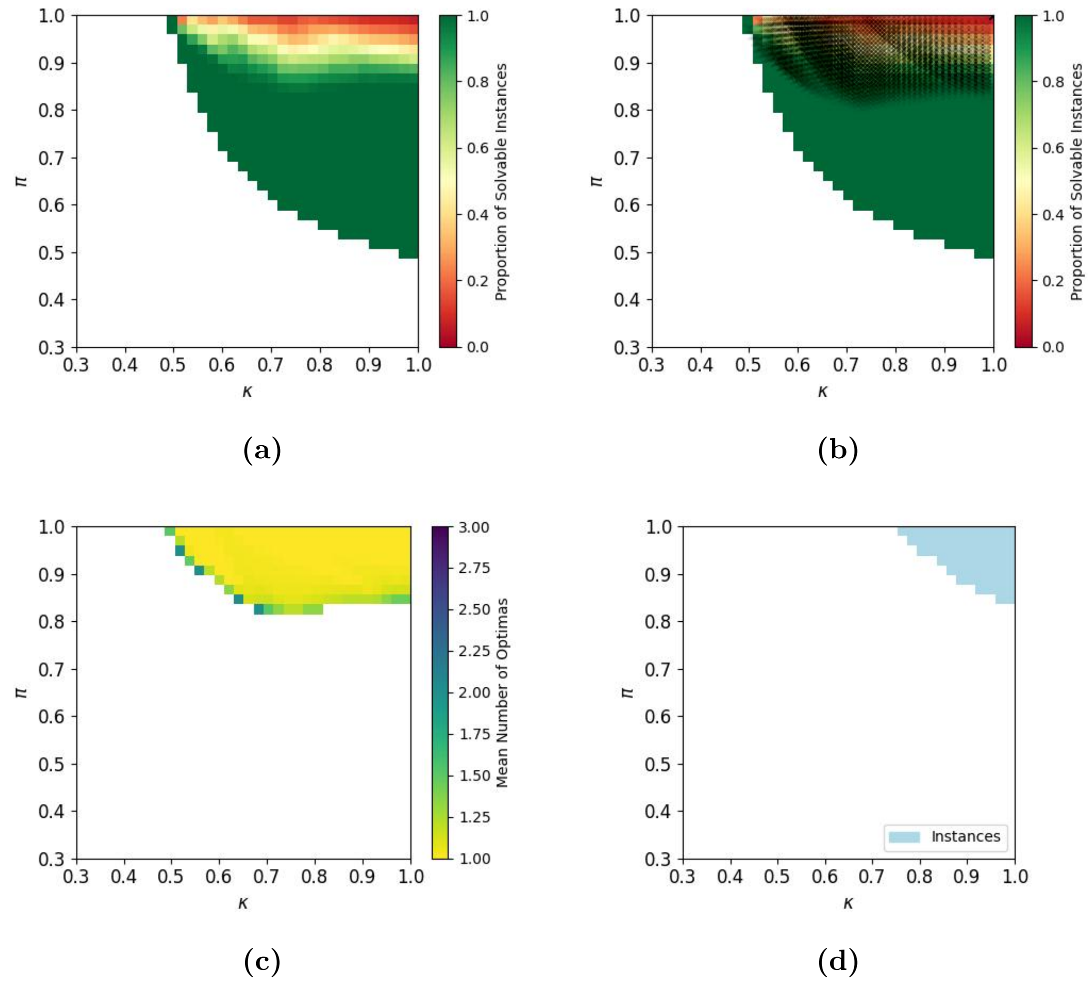
Normalized capacity and normalized profit metrics can be used to define a

[17] Treatment "WE exists" with payoff configuration "C1." See Table 2 for all treatments.

[18] The boundary of the region for which there exist instances reflects these constraints. For instance, at prices equal to payoffs, $\kappa \approx 0.48$ and $\sum_i p_i = \sum_i \phi_i$, π $(= C/(C+\sum_i \phi_i - \sum_i p_i))= 1$. So there is only 1 possible value for π when $\kappa \approx 0.48$.

[19] Feasible targets are defined as all multiples of the tick size, from a minimum equal to the budget (which can be reached without the need to trade) to a maximum equal to the total value of purchasing all goods at the lowest possible price.

[20] Unlike in past investigations of phase transitions in NP-hard problems, we do not randomly draw instances of a given κ and π. Instead, we plot the fraction of *all* instances that are solvable. In our case, the space of instances is bounded thanks to the finite price grid. There exist 3,400,320 instances in the treatment displayed in the figure. We thereby avoid biases caused by the way one would draw instances. See the discussion in Yadav et al. (2020).



Notes: $\kappa = C/\sum_i p_i$; $\pi = \Upsilon/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in Treatment "Walras – C1" in Table 2; Υ and p_i varying. (a) Solvability of decision problem instances. (b) Location of optimization problem instances (black dots) against phase transition; the location of optimization problem instances is determined by computing π with the optimum value Υ^*. (c) Location of optimization problem instances stratified by mean number of witnesses (optima); e.g., yellow locations contain instances for which the vast majority (more than 95%) have only one optimum. (d) Location of optimization problem instances only for price configurations that make all item pairs affordable, but not the triplet (necessary for equilibrium).

FIGURE 1 Location of budget problem instances in terms of normalized capacity (κ) and normalized profit (π)

difficulty metric for instances of the *optimization* version of the budget problem, as has been done for KP (Franco et al., 2021; Andrabi et al., 2026). One does so by setting the target value in the definition of the normalized profit equal to the highest attainable value:

$$\pi = \frac{\Upsilon^*}{C + \sum_i (\phi_i - p_i)} = \frac{C + \sum_i x_i^* (\phi_i - p_i)}{C + \sum_i (\phi_i - p_i)} \tag{4}$$

where x_i^* denotes the optimal allocation for item i (1 if included in the optimum; 0 otherwise). The normalized capacity κ remains defined as before.

An instance of the *budget optimization problem* is then said to be *difficult* if its κ and π (computed from Υ^*) locate it in the phase transition of the corresponding budget decision problem. Figure 1b shows the locations of all possible budget optimization problem instances, mapped on top of the solvability of the decision version, i.e., overlaid on top of Panel (a) of the same figure. We observe that most instances of the budget optimization problem are located around the phase transition. For those below the phase transition, verifying that one has reached an optimum should be easy because virtually all instances of the decision version in that region are solvable. In the phase transition, checking whether the optimum is reached is difficult because it is uncertain whether instances of the decision version at those locations are solvable, i.e., that the corresponding target values can be exceeded.

3.B Computational requirement 1: The phase transition

The first computational requirement is that prices locate the induced budget problem instance in the phase transition. Agents start with low target utility; in the experiment, they start with only cash equal to the budget C. Through trading, they will attempt to work their way up, increasing utility, and hence, increasing π. In CCE, they will have to go a long way, towards or even beyond the phase transition. As long as they are below the phase transition, it will be generically easy to find ways to improve the utility. Once entering the phase transition, the discovery of improvements is rendered far more difficult, causing those agents who prefer not to spend too much effort to stop in spite of further potential improvements. This gradual approach is precisely how humans search for a solution to the (related) 0-1 KP problem, as documented in Andrabi et al. (2026). The result is heterogeneous demand, and hence, the possibility of the market to equilibrate.

3.C Computational requirement 2: Scarcity of optima

Previous research has uncovered a second feature of instances of the KP decision problem that makes it harder for humans to correctly assess solvability, namely, the number of *witnesses* (goods bundles) that reach or exceed the target value (Franco et al., 2021). The fewer witnesses there are, the lower the accuracy in human responses. If there are only few optima to the budget optimization problem, then it will be more difficult to reach optimality. Markets can make the budget problem instance at hand harder by selecting prices for which there exists one unique optimum.[21]

It is worth recalling that, in WE, prices have to be such that there exist multiple optima for markets to equilibrate. Therefore, the *WE is not complex* when measuring complexity in terms of number of witnesses that support optimal choice.

In Figure 1c, we plot the mean number of optima of budget optimization problem instances in (κ, π) space. The same treatment is used as for Panels (a) and (b) in the figure. We observe a large overlap between the yellow region (where more than 95% of the instances have a single optimum) in Panel (c) and the phase transition region in Panel (a).

We predict that CCE will reside in the yellow region where most instances, if not all, feature unique optima. The scarcity of other solutions to the budget optimization problem instances thereby makes it less likely that humans discover the optimum, allowing those who spend most cognitive effort or are endowed with superior cognitive capability to acquire the best goods bundle, while others settle on inferior choices.

3.D Economic requirement 1: Utility-ranked equilibrium bundles

Computational complexity can generate utility-ranked demands when reaching higher-utility bundles requires greater cognitive effort or capability. This is the principle behind CCE. The other side is purely economic: demand has to equal supply. Let us now investigate the economic side.

We introduce additional notation. Let $S = (s_1, s_2, \cdots, s_I)$ denote the aggregate supply, with s_i denoting the supply for each good i. We assume $s_i < N$ for all goods, to ensure scarcity. Let $x^n = (x_1^n, x_2^n, \cdots, x_I^n)$ with $x_i^n \in \{0, 1\}$ denote

[21] Multiple optima can be allowed only if they can be reached by different algorithms with the same amount of effort (number of computations). The example of Section 2 assumes that agents choose one of the Sahni-k algorithms. There, the number of computations increases strictly in k. In addition, the algorithms are deterministic, implying that they will always estimate the same optimum for a given instance. Consequently, there does not exist an instance with two or more optima that can be reached with equal number of computations by two different algorithms.

the individual demand of agent n for good i. The aggregate demand is denoted as $X = \sum_n x^n$. An equilibrium bundle is a subset of the goods that will have to be held in equilibrium by some agent(s). Let $d^j = (d_1^j, d_2^j, \cdots, d_I^j)$ with $d_i^j \in \{0, 1\}$ denote equilibrium bundle j, and denote by $\mathcal{D}$ the set of all equilibrium bundles $d^j, j = 1, ..., J$, where J equals the number of equilibrium bundles. For readability, we will use symbols instead of binary numbers. For example, a bundle $d = (1, 0, 1)$ is equivalent to $d = \{L, H\}$ and $d = \{H, L\}$. Equilibrium requires markets to clear, such that $X = S$. Below we describe the properties that our equilibrium bundles need to satisfy.

CCE requires that the equilibrium bundles are utility ranked. Without loss of generality, we denote this ranking using an index j such that $U(d^j) < U(d^{j+1})$; in other words, bundle d^{j+1} yields higher utility than bundle d^j. The utility ranking ensures that cognitive effort or capability to reach higher utility is compensated. Here, we use strict inequalities, though it could be that two different choices lead to the same utility level and with the same effort.[22] Sticking to strict inequalities, we have a simple necessary requirement for CCE to exist: prices have to be such that there exists a unique optimum.

3.E Economic requirement 2: Affordability and market clearing

Our second economic requirement combines affordability with market clearing. Let y^j represent the mass of agents who choose equilibrium bundle d^j. We require that, for each agent n, there exists some equilibrium bundle d^j such that $x^n = d^j$. So, $y^j > 0$. Market clearing then imposes that the weighted aggregate demand (D) must satisfy

$$D \equiv N \times \sum_j y^j d^j = S.$$

Informally, this means that all agents only select an equilibrium bundle, in appropriate weights so that the weighted sum of the demanded bundles matches the aggregate supply.[23]

[22]In Section 2, this could not happen because of the nature of the algorithms agents use.

[23]To illustrate, take our motivating example in Section 2. There are three equilibrium bundles, namely, $d^1 = \{M, L\}$, $d^2 = \{H, L\}$, and $d^3 = \{H, M\}$. When the budget is 3,200, the bundles yield utilities equal to 5,000, 5,200, and 5,600 respectively. When the budget is 2,800, they yield utilities equal to 4,800, 5,000, and 5,400 respectively. Thus, they are always utility ranked as their corresponding utilities are increasing. Under the assumption that the agents use algorithms in the Sahni-k class, then the utilities correspond to solutions from k=2, k=1, and k=0 respectively. As such, utility rankings correspond to the ranking based on the diffi-

Choices have implications for prices, and hence, for the location of the budget problem instance in (κ, π) space. In particular, prices underlying κ and π have to be such that all equilibrium bundles are affordable, but not all the goods simultaneously. In Figure 1d, we plot the location of budget optimization instances (based on π evaluated at the utility for the best bundle only) for price configurations for which all equilibrium bundles are affordable, but not the triplet (here, there are three goods). The same treatment is used as for Panels (a), (b) and (c) of the figure.

For the reader to gain perspective, we note that there are 25,760 instances depicted in Panel (c) (where all optimization instances are plotted, regardless of whether they satisfy economic requirements), while in Panel (d), there are only 3,695 instances. Our second economic requirement alone reduces by 85 percent the number of potential budget optimization problem instances that the market can choose from.

4 Experimental design and predictions

The experiment is designed with the numerical example of Section 2 as inspiration. On the demand side, we induce preferences as in the example. The goods traded are claims to a deterministic monetary payoff (hence, they are risk-free). Because preferences are induced monetarily, a consumer's utility equals their period earnings, and we use the two terms interchangeably. On the supply side, we introduce sellers with zero marginal costs. We let markets determine prices and allocations. We organize exchange through a continuous double-sided open limit-order book system. Among agents, we distinguish between consumers and sellers.

4.A The market and the trading game

In each experimental session, participants trade in a market for 16 equal-duration periods. There are three goods available for trade with pre-determined payoffs to consumers only. At the end of each period, goods are liquidated and participant period earnings are calculated based on final liquidated holdings. All periods are independent; one's final holdings and actions in one period do not influence any of the other periods. One quarter of the participants are sellers, and three

culty of finding them. For market clearing, we require exactly one third of agents to select each demand bundle $\left(y^1 = y^2 = y^3 = \frac{1}{3}\right)$. In that case, the weighted sum of equilibrium demands is $D = 3 \times \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right) = (2, 2, 2)$, which is equal to the aggregate supply of two units of each good for every three agents.

quarters are consumers. The sellers act as suppliers of goods. Each seller is endowed with two units of each of the three goods and no cash. They do not have a claim to the payoffs on the goods, so they are induced to sell the goods for as much cash as possible. The consumers represent the demand side of the economy. Each consumer is endowed with cash and no goods. The consumers are only compensated for the *first unit* of each good they hold at the end of the period, and for any remaining cash.

A noteworthy feature of our incentive scheme is that all goods are initially in the hands of sellers who have zero marginal cost, and all cash is in the hands of consumers who all receive the same payoffs on the goods, and hence, who are homogeneous. This feature serves multiple purposes. First, it ensures that the decision of which good to purchase is difficult for consumers while keeping the complexity of the supply side minimal. Second, it maximizes trade, since it is in everyone's interest (i.e., Pareto-improving) that all goods move from sellers to consumers. The large expected number of trades increases the power of our design.[24] Third, it creates price dynamics that stack the deck against our equilibrium. CCE requires high prices so that consumers can only afford pairs of goods (see Economic Requirement 2). With divisible goods, it is known that inducing sellers with zero marginal cost causes prices to end up far below equilibrium levels (Smith and Williams, 1982; Rasooly, 2022). If such forces are at play for indivisible goods as well, then they must push prices away from CCE. These forces would destroy any evidence in favor of CCE.

The trading protocol is an online open-book continuous-time double-sided auction.[25] Traders submit limit orders, i.e., the maximum price they are willing to pay for buying one or more units of a good or the minimum price they are willing to accept for selling one or more units of a good. A trade occurs in two scenarios: (i) a buy order is submitted with a price higher than or equal to the lowest standing selling price, or (ii) a sell order is submitted with a price lower or equal to the highest standing buying price. If a trade occurs, the price is determined by the best standing order at the moment a trading order is submitted. In the

[24]Contrast this with some of the designs in Plott and Sunder (1982, 1988): assuming risk neutrality, we do not expect a single trade, so in expectation we have no observations with which to verify whether prices are right. The same issue emerges in the design of Smith et al. (1988).

[25]The software, Flex-E-Markets, is used on a subscription basis. See quantahm.com. Most major stock exchanges, such as the New York Stock Exchange, the London Stock Exchange, Euronext, and NASDAQ, operate with this trading protocol. Settlement in Flex-E-Markets is immediate, however, and therefore part of the clearing mechanism. The latter wputs additional computational constraints on the software compared to exchanges in the field.

first scenario, the trade is triggered by a consumer submitting a buy order, so the standing sell order (ask) sets the price. In the second scenario, the trade is triggered by a seller submitting a sell order, so the standing buy order (bid) determines the price. If the buy and sell orders are for different quantities, the order with the higher number of units is split and the remaining units (after trade) are converted into a new order at the original order price and the new order is then subject to the same trade evaluation process. If trade does not occur, the order becomes a standing order and is anonymously displayed to all other traders, ready to trade with later orders. Standing orders can be canceled by the submitter prior to trading and are removed at the end of each period.

The trading interface consists of three parts: the order book, the order form, and the holdings account. A snapshot of the trading interface can be found in Appendix G.

The order book consists of three panels, one for each of the goods. Each good-panel updates in real time and stores all standing orders with their price and corresponding quantity. The standing orders are anonymized and participants can only identify their own standing orders. The orders are priority-ranked. Buy orders are shown in descending order and sell orders in ascending order. For orders with the same price, earlier orders receive priority and execute first, however quantities are combined in the interface to show the total standing volume at a given price. Participants also see the (anonymized) trade history of all trades that occurred for each good in chronological order. Color coding ensures that the trader can determine whether the trade was triggered by an incoming buy order (blue) or an incoming sell order (red).

The order form is the only way participants can communicate with the market and its participants. No other communication between participants is allowed. Participants can submit buys and sells for any or all of the three goods. The submitted prices have to be positive and cannot exceed a maximum price.[26] A tick size of 0.05 is imposed. This tick size was chosen, among others, so that, in the two treatments where WE exists, WE prices are unique.

The holdings account displays for each participant their settled and available goods and cash. "Settled" holdings represent the participant's *de facto* current holdings. Placing an order into the market does not affect these claims until the order is traded. When a trade occurs, the settled balance updates to reflect that trade (good increase and cash decrease for a traded buy order, and good decrease

[26]Maximum prices for all goods were set equal to 25% above the highest payoff of all three goods, rounded up to the nearest experimental currency unit.

and cash increase for a traded sell order). At the end of the period, liquidation is based on the settled holdings at that time. "Available" holdings act as a real-time budget constraint on participants. They equal the settled holdings minus any goods or cash committed to active standing orders. For example, if a seller has 2 units of H and places a sell order for 1 unit, that unit becomes committed and unavailable for other orders, leaving an available balance of 1. The seller can cancel the standing order to free the unit for a different price. Similarly, when a consumer places a buy order, the cash offered is committed, reducing their available cash by the buy order amount. Available holdings adjust whenever orders are placed or canceled. Available holdings also incorporate shorting limits, though in our experiment short sales and cash borrowing are not permitted.

4.B Experimental treatments

Our experiment implements a mixed 2×2 design with four treatments. One treatment variable is manipulated between participants, and one is manipulated within participants. Table 2 presents a concise overview of all our experimental treatments.

TABLE 2 Overview of experimental treatments in terms of cash (budget) and payoffs for consumers

Budget Treatment	Payoff Treatment	Cash C (Budget)	Good Payoff (ϕ_i) L	M	H	Total sessions	Total participants
No Walras	C1	2.80	1.20	2.40	3.00	5	88
	C2	2.20	1.60	2.40	3.20	5	88
Walras	C1	3.20	1.20	2.40	3.00	5	92
	C2	2.60	1.60	2.40	3.20	5	92

In the first two sessions of the "No-Walras" treatment, the payoffs for all goods were one fourth of the values presented here; there, we used a conversion rate of experimental currency into pounds of 1:4. The payoffs in all other sessions were as displayed above, and a conversion rate of 1:1 was used.

Across sessions, and hence *across participants*, we vary the cash endowment to participants acting as consumers. In one half of our sessions, referred to as "No Walras" sessions or Sessions S1 to S5, consumers are endowed with cash that is insufficient for WE to exist. In the remaining sessions, called "Walras" sessions or Sessions S6 to S10, the cash endowment is sufficient for WE to exist.

Within participants, we vary the goods payoffs to the consumers. We refer to the two payoff configurations as "C1" and "C2." In each session, one block of eight periods uses payoff configuration C1, and a second block uses payoff configuration C2. The order of the blocks is counterbalanced between sessions. We vary the goods payoffs as a robustness check to ensure that our results are not driven by a specific payoff parametrization. Note that while we refer to the goods as H, M and L here, in the experiment they were called A, B and C to prevent any biased expectations by the participants, and the letter assignment was shuffled between payoff configurations.

Within payoff configurations, we also vary roles (consumer; seller), so that all participants are exposed to both sides of the economy. One in four periods, a participant is a seller; in the remaining periods, the participant acts as consumer. The rotation is implemented to induce fairness and to ensure that consumers understand the incentives for sellers and *vice versa*.

4.C Predictions

This section motivates several testable predictions that follow from our theory. The predictions are presented informally, while formal statistical tests for the predictions are discussed along the results.

We first present predictions that need to be fulfilled to satisfy the economic requirements for CCE listed in Section 3. Subsequently, we deal with predictions that flow from the computational requirements for CCE. Finally, we add predictions that would obtain only for the "Walras" Treatment since they pertain to the WE.

4.C.1 Economic predictions

Because sellers have zero marginal cost and consumers assign positive value to the entire supply of available goods, all goods need to be sold. As such, the resulting reallocation exhibits high efficiency. From the consumer side, due to the ratio of potential aggregate supply and demand, market clearing would require an equal share of consumers to purchase each of the three pairs of goods (see Economic Requirement 2).

Three predictions follow. They do not depend on the level of cash (budget) allocated to the consumers, so obtain in both the Walras and No Walras treatments.

Prediction 1. *Efficiency is 100%: all units of goods are sold to the consumers.*

Prediction 2. *All consumers end up holding pairs of goods.*

Prediction 3. *Exactly the same number of consumers end up holding each pair of goods.*

As to prices, consumers should be able to afford two but not three goods. Among others, this implies higher prices when budgets are higher (as in the Walras Treatment). Therefore, we formulate the following predictions.

Prediction 4. *Prices are sufficiently high so that any pair but not all three goods can be afforded.*

Prediction 5. *Prices are higher when consumers have larger budgets.*

So far, our predictions are required for any generic equilibrium to hold. However, one more restriction on prices needs to be imposed. CCE requires that the equilibrium pairs entail different utility levels, so that cognitive effort expended or cognitive capability needed to reach the optimum are compensated.

Prediction 6. *The three equilibrium goods pairs earn different levels of utility.*

4.C.2 Computational predictions

We now present predictions regarding instance complexity. The first prediction is driven by the requirement that the computational complexity of the consumers' budget problem instances needs to be high. As discussed earlier, we interpret this to mean that the market selects prices so that the equilibrium instance is located in the phase transition. This is a restatement of Computational Requirement 1.

Prediction 7. *Prices set the budget problem instance into the region of the phase transition.*

Since prices are determined endogenously as orders arrive in the market, they (prices) may occasionally move the budget problem instance out of the phase transition. There, the budget problem instance tends to be easy, and hence, with little effort all consumers may find the optimum. Demands are no longer heterogeneous, so markets cannot equilibrate. Prices must push the budget problem instance back into the region of the phase transition.

Prediction 8. *Prices mean-revert to locate the budget problem instance inside the phase transition.*

For humans, the chance of finding the optimum in allocation problems such as KP increases when there are multiple optima (multiple "witnesses"). This means that complexity, to humans at least, rises when there are fewer optima. We referred to this as Computational Requirement 2. It leads to the following prediction.

Prediction 9. *Prices are such that the budget problem instance, at optimal utility, is in the region where most if not all possible instances feature a unique optimum.*

This prediction is closely related to Prediction 6. When prices are such that the budget problem instance has a unique optimum (Prediction 9), it must be that at least two of the equilibrium goods pairs generate different utility levels. Conversely, if utility levels of all equilibrium pairs are the same, contrary to Prediction 6, Prediction 9 cannot be right, unless nobody optimizes.

4.C.3 Walrasian equilibrium

We also aim to test CCE against the traditional WE when it exists, as is the case in the "Walras" Treatment. In that case, the lower complexity of WE (it is outside the phase transition and it entails a budget optimization problem instance with three optima) favors its emergence. However, this assumes that the market has already settled on WE prices. Since there are more price configurations that are consistent with its necessary conditions, CCE may be obtained instead. We formulate a hypothesis that can be tested by investigating where prices appear to converge.

Prediction 10. *Goods prices converge to levels that are significantly different from WE prices.*

Another forceful way to reject WE is to find that equilibrium pairs all gain different levels of utility. That is, WE is falsified if the data support Prediction 6.

4.D Implementation

We ran ten in-person experimental sessions in total. The sessions took place at the Cambridge Experimental and Behavioural Economics Group (CEBEG) Laboratory of the Judge Business School (Cambridge), between March and November of 2024. All participants were recruited from the CEBEG participant pool which is open to the general population of 18 years or older and able to attend in-person experiments. Ethical approval was obtained prior to data collection, which included the requirement of informed consent.[27]

[27]Ethics Approval UCAM-FOE-24-02 (University of Cambridge, 2024).

Experiment sessions lasted three hours. Participants were assigned randomly to a computer station. Before the experiment began, informed consent was obtained from all participants. We collected basic demographic data (age, gender, and field of study). Experiment instructions were read out by the experimenters and a demonstration of the trading interface was given, to ensure common knowledge of the rules of the experiment and the trading interface.[28] The instructions and demonstration of the trading interface lasted approximately one hour. The instructions included numerical examples of how period earnings and take-home pay were calculated. We ran two non-timed practice periods for participants to further familiarize themselves with the trading interface. We also ran four practice periods, with the same duration as the experiment periods (namely, three minutes), and where participants were moved from one to the other side of the market across periods. When all clarifying questions were answered, a short break was introduced. The 16 experimental periods lasted about one hour, after which the participants were paid in cash, and the experiment concluded.

Roles were counterbalanced so that each participant was a consumer in 12 periods and a seller in four periods, while three out of four participants in any given period were consumers. For the latter to obtain, we needed the number of participants for each session to be a multiple of four. We aimed to run each session with 20 participants, though some sessions ran with 16. Any participant in excess of 20 (or 16 for some sessions) was sent away with a show-up reward of £10. Participants were provided with an additional sheet of paper that they were instructed to use to record their role at the beginning of each period, to avoid confusion.

The participants knew their own role and their own endowment, but no information about other participants' roles and endowments was provided. This information structure is typically assumed in general equilibrium theory, and hence, a well-designed general equilibrium experiment should adhere to it. In principle, only trade prices are to be made available to participants, but the theory assumes that these are equilibrium prices, while the theory does not explain how they would come about. Therefore, general equilibrium experiments with continuous double-sided auctions usually allow access also to all standing (active) limit orders, i.e., the book of orders is made open. This approach has been successful in generating market equilibrium in past experiments, starting with Smith (1965).

Participants were informed at the beginning of the session that they would be paid for four randomly drawn periods out of the 16 experimental periods. The

[28]Instructions are reproduced in Appendix F.

paid periods were drawn at the end of each experimental session, to ensure high incentives to perform well throughout all periods. The drawing ensured that all participants were paid thrice as a consumer and once as a seller.

In total, we engaged 180 participants. They were on average 26 years old (median = 24, sd = 7.76, min = 18, max = 60), and evenly balanced across genders. Each participant was allowed to join only one session. They earned on average £35.50 (median = £36.00, sd = £2.75, min = £27.00, max = £42.00). An overview of demographic characteristics across treatments is provided in Table 3.

TABLE 3 Participant demographics

Budget Treatment	Total sessions	Total participants	Earnings	Age	Gender			
					Male	Female	Other	Unknown
No Walras	5	88	£35.25	26	48	39	0	1
Walras	5	92	£35.75	26	42	45	3	2

For earnings and age, we report the mean. For gender, we report the number of participants identifying as male, female, other (non-binary or unlisted gender), or unknown (prefer not to say).

5 Results

We group our results into three subsections. The reader may want to inspect Appendix E to get an appreciation for the trading activity that emerged in our experiment. One of the features that stands out is the volatility of prices. We come back to this later.

5.A Economic requirements

We begin with Table 4, which presents the goods allocations of consumers at the close (end) of each period and the efficiency of goods sales.

Prediction 1. Since sellers face zero marginal cost, they should be willing to sell goods at any price. Column 3 shows that roughly 90% of goods move from the sellers to the consumers in all treatments. Turning this around, 10% of goods are mistakenly kept by sellers. Consumers do occasionally make mistakes too: a small percentage of them (4.5% of consumer-periods) end up buying more than one unit of the same good, even though they gain nothing from the excess units (and pay a positive price for them). There is a reason not to consider over-buying as "mistakes," since it could reflect speculation, when consumers purchase additional

TABLE 4 Final allocations of goods per consumer-period

Budget Treatment	Payoff Treatment	Efficiency	Triplet $\{H,M,L\}$	Pairs $\{M,L\}$	$\{H,L\}$	$\{H,M\}$	Singletons $\{L\}$	$\{M\}$	$\{H\}$	Zero Goods	Excess Units	Total
No Walras	C1	93.5%	32	94	127	143	21	32	26	24	29	528
	C2	89.9%	19	115	130	130	19	30	51	19	15	528
Walras	C1	89.4%	10	111	147	124	20	38	49	24	29	552
	C2	86.9%	13	143	119	98	13	27	98	17	24	552
Aggregate		89.9%	74	463	523	495	73	127	224	84	97	2,160

Efficiency is calculated as the percentage supply (units of goods) that consumers acquired from sellers by the end of a period. Unit of observation is "consumer-period," that is, the final allocation of one consumer at the end of one period. "Excess Units" refers to the number of consumer-periods when the consumer held at least one good in excess of one unit (for which they were not compensated). Consumer-periods when units are held in excess do not count towards the consumer-periods of the triplet, pairs, or singletons. A more granular breakdown of final allocations including breakdown of excess demands is provided in Appendix D.

units in order to sell later for a profit.[29] In general, the data support Prediction 1.

Prediction 2. We observe that most consumers end up with a pair of goods. This happens in approximately two thirds (1,481/2,160 or 68.6%) of the consumer-periods. In equilibrium, this percentage should have been 100%, but the high volatility in prices occasionally allows a small number of consumers to purchase all three goods. Similarly, roughly one-fifth of consumers buy only a single good, and an even smaller fraction do not purchase any goods at all. Therefore, the evidence mostly supports Prediction 2.

Prediction 3. Among consumers who purchase pairs of goods, the proportion choosing each possible pair is approximately the same. In total, consumers obtain the pair $\{M,L\}$ in 463 consumer-periods, $\{H,L\}$ in 523 consumer-periods, and $\{H,M\}$ in 495 consumer-periods. We formally test the equality of proportions by first estimating a multinomial logistic model with random intercepts at the consumer and session level, and then testing whether the estimated proportions differ. We fail to reject the hypothesis that all three pairs are chosen with equal likelihood ($\chi^2(2) = 3.65$, $p = 0.1614$, $N = 1,481$). Thus, the data support Prediction 3.

We summarize the findings so far as follows.

Interim Conclusion 1. *The data support predictions 1 to 3.*

Prediction 4. Prices are expected to be sufficiently high so that consumers

[29] The idea that excess purchases reveal speculation is reinforced by the finding that almost an equal number of consumer-periods (78, versus 97) manages to sell excess units before the end of the period. Therefore, we could claim that in total 78+97 = 175 attempts at speculation are tried, of which a bit less than half are successful. So, out of a total of 2,160 about 8% of consumer-periods could be classified as involving speculation.

cannot afford all three goods. Table 5 lists the frequencies with which participants could afford all three goods. We calculate the frequency as a fraction of total period duration that all three goods could be afforded. Duration is measured as either calendar time or trade time. Under the second method, one time unit corresponds to a trade. For prices, we take either the last traded prices[30] or the best standing sell price for the three goods.

We observe that participants could very rarely afford all three goods. This is even more pronounced for the standing sell (ask) prices, which are, as expected, higher than the trading prices.

TABLE 5 Fractions of time when goods triplet is affordable

Budget Treatment	Payoff Treatment	Last Traded Price Calendar Time	Last Traded Price Trade Time	Best Standing Sell Calendar Time	Best Standing Sell Trade Time
No Walras	C1	6.0%	6.8%	0.0%	0.1%
	C2	5.5%	6.8%	0.0%	0.0%
Walras	C1	1.3%	1.9%	0.0%	0.0%
	C2	4.6%	5.2%	0.1%	0.1%
Aggregate		4.3%	5.2%	0.0%	0.0%

Table presents fractions of time when the goods triplet $\{H, M, L\}$ is affordable. Time is measured as calendar time or as count of trades ("Trade Time"). Affordability is based on either transaction prices of most recent trades ("Last Traded Price") or from best sell offer in the book ("Best Standing Sell"). Affordability is only calculated from when all three goods have a price, that is, after each good has traded at least once in the period.

However, Table 6 shows that often prices are too high, so that one or more pairs become unaffordable, contrary to our prediction. In the table, we only report results for affordability at traded prices. Time is measured as count of trades.

Prediction 5. In Appendix E we provide time series plots of the raw trade prices (Figure A5), and of the prices of pairs and the triplet of goods (Figure A6). Those graphs visualize the evolution of prices. The observation that prices are uniformly higher when the budget is higher (i.e., when WE exists) is immediately apparent from these plots and does not require a formal test.

Interim Conclusion 2. *The data support Prediction 5. Support for Prediction 4 is qualified, however: while prices are sufficiently high so that all three goods could*

[30] At any point in time, only one good trades; in those cases, prices for the non-traded goods are set equal to previous (last) trade prices, as is convention in financial economics.

TABLE 6 Fractions of time when one of the equilibrium goods pairs is NOT affordable

Budget Treatment	Payoff Treatment	Goods Pairs $\{M, L\}$	$\{H, L\}$	$\{H, M\}$	At Least One Pair
No Walras	C1	0.2%	3.5%	43.3%	43.3%
	C2	3.6%	26.2%	47.3%	49.8%
Walras	C1	0.1%	9.7%	69.7%	69.8%
	C2	3.6%	41.5%	69.4%	71.0%
Aggregate		1.8%	19.7%	57.1%	58.1%

Table presents affordability of pairs of goods. Affordability is defined as the fraction of time (measured as count of trades, i.e., trade time) in a period when the pair at the top of the column is not affordable, or when at least one pair is not affordable (rightmost column). Affordability is only calculated from when all three goods have a price, that is, after each good has traded at least once in the period.

not be afforded together, between 1/2 and 2/3 of the time, not all pairs are affordable.

Prediction 6. The last key economic prediction of our equilibrium is that the various equilibrium goods pairs ought to yield different utilities. To test this prediction, we estimate the long-run mean differences in total earnings from purchasing the pairs $\{H, M\}$ and $\{H, L\}$, as well as from purchasing $\{H, L\}$ and $\{M, L\}$.

We expect both of these differences to be significantly different from zero. We do not know whether the differences ought to be strictly *positive*. In the numerical example of Section 2, they are. But we should allow for the possibility that, say, the first difference $(U_t(\{H, M\}) - U_t(\{H, L\}))$ is positive while the second one $(U_t(\{H, L\}) - U_t(\{M, L\}))$ is negative. This would happen if the pair $\{H, L\}$ generated lowest utility, while the pair $\{M, L\}$ generated medium utility. In that case, the expectation of the absolute value of the second difference in utility should be smaller than the expectation of the first difference.

The evolution of the two differences is estimated using a vector autoregression model (VAR). The unit of observation is a trade and one VAR is estimated per session–payoff treatment, concatenating the eight periods of the treatment.[31]

[31]We ignored changes in the two differences that straddled period ends. We also estimated VARs for each period separately and then aggregated the results per session-treatment. This led to qualitatively the same inference, but because we computed standard errors from the cross-section of estimated long-run means, we encountered a large loss in power.

Formally, we estimate the following VAR:

$$\begin{bmatrix} U_t(\{H,M\}) - U_t(\{H,L\}) \\ U_t(\{H,L\}) - U_t(\{M,L\}) \end{bmatrix} = \mu + \sum_{k=1}^{K} \Phi_k \begin{bmatrix} U_{t-k}(\{H,M\}) - U_{t-k}(\{H,L\}) \\ U_{t-k}(\{H,L\}) - U_{t-k}(\{M,L\}) \end{bmatrix} + \epsilon_t. \tag{5}$$

We then test whether the long-run expectations of the differences in values are significantly different from zero, i.e., whether elements of the vector $(I - \Phi_1 - \cdots - \Phi_K)^{-1}\mu$ are nonzero.[32] Inspection of the autocorrelations of the error terms consistently suggests that the most parsimonious VAR uses only one lag, i.e., $K = 1$ in Equation 5. Likewise, the estimates of the autoregression coefficient matrix (Φ_1) confirm that the time series are stationary, i.e., that the earnings differences are mean-reverting. The results of the estimation are displayed in Table 7.

TABLE 7 Estimated long-run earnings differences between goods pairs

Budget Treatment	Payoff Treatment	Session 1	2	3	4	5
No Walras	C1	71***	49***	73***	65***	84***
	C2	65***	59***	48***	67***	63***
Walras	C1	68***	43***	43***	66***	65***
	C2	45***	56***	62***	68***	67***

(a)

Budget treatment	Payoff treatment	Session 1	2	3	4	5
No Walras	C1	6	38***	47***	8	9*
	C2	14***	41***	57***	15***	-2*
Walras	C1	-19**	16**	15***	-3	7
	C2	13	37**	26***	-18***	-13***

(b)

(a) Earnings differences between $\{H,M\}$ and $\{H,L\}$, in 0.01.
(b) Earnings differences between $\{H,L\}$ and $\{M,L\}$, in 0.01.
Legend: ***: $p \leq 0.001$, **: $p \leq 0.01$, *: $p \leq 0.05$, all two-sided.

The results confirm that the long-run expectations of earnings levels are generally significantly different across equilibrium goods pairs. The first difference $(U_t(\{H,M\}) - U_t(\{H,L\}))$ is uniformly positive and significant. That is, the

[32]Standard errors of the estimated long-run means are computed from the standard errors of the estimated coefficients using the Delta Method (see, e.g., Schervish (2012), Section 7.1.3).

rankings of the utility earned from buying $\{H, M\}$ and $\{H, L\}$ is as in the numerical example in Section 2. As to the second difference $(U_t(\{H, L\}) - U_t(\{M, L\}))$, we observe that the long-run expectation is significantly ($p \leq 0.05$) positive in the majority (11) of the 20 session-treatments, but it is significantly ($p \leq 0.05$) negative in four session-treatments. When negative, the magnitude (absolute value) of the long-run expectation of the second difference is never more than 1/3 of that of the first difference. In the remaining 5 sessions, the difference in the expected long-run values of the pairs $\{H, L\}$ and $\{M, L\}$ is not significant. This is consistent with CCE only if reaching the utility level of these two pairs requires equal effort or ability.

The utility differences of pairs $\{H, M\}$ and $\{H, L\}$ are also economically significant. For illustration, an estimated coefficient of 71 (Treatment: No Walras – C1) corresponds to earnings of £0.71 per period. The fact that higher compensation is provided for moving from the second-best to the first-best bundle, compared with the compensation for moving from the third-best to the second-best bundle, suggests that, if cognitive effort spent is the driving force behind our observations, the cognitive effort cost associated with discovering better bundles is strictly convex.

TABLE 8 Average earnings implied from final period holdings including cash

Budget Treatment	Payoff Treatment	Final Goods Holdings $\{M, L\}$	$\{H, L\}$	$\{H, M\}$
No Walras	C1	£4.19	£4.50	£5.50
	C2	£4.44	£4.90	£5.64
Walras	C1	£4.42	£4.49	£5.44
	C2	£4.57	£4.88	£5.64

Only participant-periods for which the participant holds one of the three equilibrium pairs (shown at top of the columns) are included. In all cases (rows), the Kruskal-Wallis H test rejects equality of the distributions of earnings across goods pairs holdings ($p < 10^{-4}$).

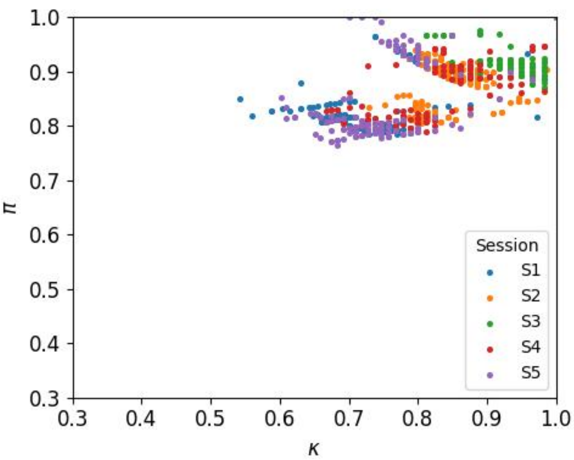
Table 8 concludes the same in a different way. It reports the average earnings (in pounds, including cash) at the end of a period of consumers who hold the goods pair shown at the top of each column. All consumers start with the same cash, but could spend part of it on acquiring the pair. In CCE, the average earnings across pairs should be significantly different. In the WE, when it exists, average earnings across pairs should be the same. We observe that average earnings for $\{H, M\}$ are the highest, followed by those for $\{H, L\}$, and lowest for $\{M, L\}$.

Because the distributions of earnings categorized by final goods pair holdings are distinctly non-Gaussian, we resort to a non-parametric test to formally confirm the findings, namely, the Kruskal-Wallis H statistic. This tests whether the nature of final holdings has no effect on the earnings in the sense that there is no order among them. The test rejects when there is order: holding of one pair tends to give the lowest earnings, another gives the next lowest earnings, and so forth. Technically, it investigates stochastic dominance ordering. In all treatments (rows), we reject equality of earnings distributions at high significance levels ($p < 10^{-5}$). This is contrary to the WE, where earnings distributions should not exhibit stochastic dominance regardless of the pair a participant chooses to hold.[33]

Interim Conclusion 3. *On balance, the data support Prediction 6.*

5.B Computational complexity

Prediction 7 and Prediction 8. The first Computational Requirement states that prices locate the budget problem instance inside the phase transition in (κ, π) space, and when not, that there is a strong tendency for the κ and π to mean-revert to the phase transition.



Treatment "No Walras, C1." $\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i, p_i, and Υ^* as in shown treatment. Each dot corresponds to a trade. the remaining three treatments are qualitatively similar and are shown in Figure A1 in Appendix C.

FIGURE 2 Recorded locations of the budget problem instance in $\kappa \times \pi$ space, representative treatment.

For the reader to get perspective on the formal statistical analysis to follow, in

[33]Dunn's test with Bonferroni correction rejects pairwise equality across final goods holdings categories at $p < 0.01$, except when comparing $\{M, L\}$ and $\{H, L\}$ in treatment "Walras – C1."

Figure 2 we display the observed location of the budget problem instances in (κ, π) space after each trade for a representative treatment.[34] κ and π are computed based on traded prices and the optimum value Υ^* of the budget problem instances they induced. The four experimental treatments are qualitatively similar; all four are shown in Appendix C. Sessions (S#) are color-coded. Realized κs and πs cover a large space, going beyond the phase transition (see Figure 1a). This is acceptable, provided the evolution of κs and πs shows a strong tendency to mean-revert to the phase transition.

To formally validate mean-reversion to the phase transition region, we fit a VAR model to the time series data on which Figure 2 is based. That is, we estimate the following model:

$$\begin{bmatrix} \kappa_t \\ \pi_t \end{bmatrix} = \mu + \sum_{k=1}^{K} \Phi_k \begin{bmatrix} \kappa_{t-k} \\ \pi_{t-k} \end{bmatrix} + \epsilon_t. \qquad (6)$$

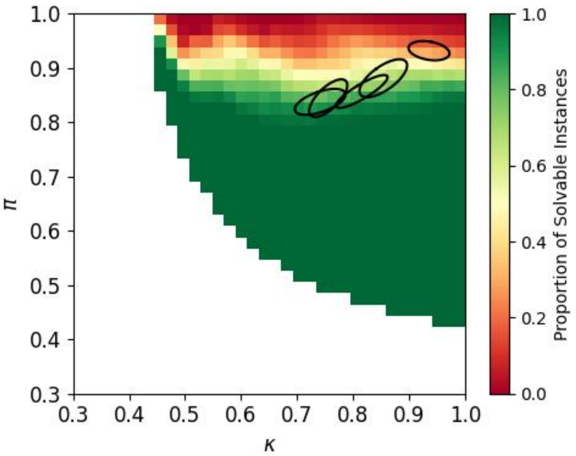
As we did with the differences in utilities from pairs of goods, we concatenate the data for all periods within a session–payoff treatment and estimate the parameters. We derive an estimate of the long-run mean by computing the vector $(I - \Phi_1 - \cdots - \Phi_K)^{-1}\mu$. Inspection of the error autocorrelations suggests that the best parsimonious model uses only one lag of the observed time series. Estimates of the roots of the autoregression (not reported) indicate strong mean reversion in the two time series, which confirms Prediction 8. The estimates of the long-run means themselves are listed in Table 9.

TABLE 9 Estimates of long-run expectation of normalized capacity κ and normalized profit π.

Budget Treatment	Payoff Treatment	(κ, π) Session 1	2	3	4	5
No Walras	C1	(0.74, 0.84)	(0.85, 0.88)	(0.94, 0.93)	(0.82, 0.86)	(0.76, 0.84)
	C2	(0.76, 0.80)	(0.77, 0.82)	(0.85, 0.86)	(0.76, 0.80)	(0.77, 0.79)
Walras	C1	(0.76, 0.89)	(0.80, 0.89)	(0.82, 0.86)	(0.72, 0.88)	(0.74, 0.87)
	C2	(0.76, 0.80)	(0.78, 0.85)	(0.71, 0.80)	(0.72, 0.81)	(0.75, 0.81)

We subsequently compute estimates of the standard error of the long-run mean of the VAR in Equation 6. From these, we calculate 95% confidence regions. Finally, we superimpose them on the phase transition region for each treatment

[34]Since only one good trades each time, we have to choose prices for the non-traded goods. As before, we take those to be the prices at which the goods last traded. Observations are only plotted after each good has a last trade price, that is, after it has traded during the period.



Treatment "No Walras, C1." $\kappa = C/\sum_i p_i$; $\pi = \Upsilon/(C+\sum_i(\phi_i - p_i))$; C, ϕ_i as in shown treatment; p_i, and Υ varying; 95% confidence ellipsoid based on Equation 6, where p_i equals trade price and Υ is value at the optimum choice. Phase transitions are indicated in light green down to orange. The remaining three treatments are qualitatively similar and are shown in Figure A2 in Appendix C.

FIGURE 3 95% confidence ellipsoid of estimated asymptotic values for κ and π, relative to the phase transition map of the budget problem, representative treatment.

separately. The phase transition regions are obtained analogously to the one depicted in Figure 1a.

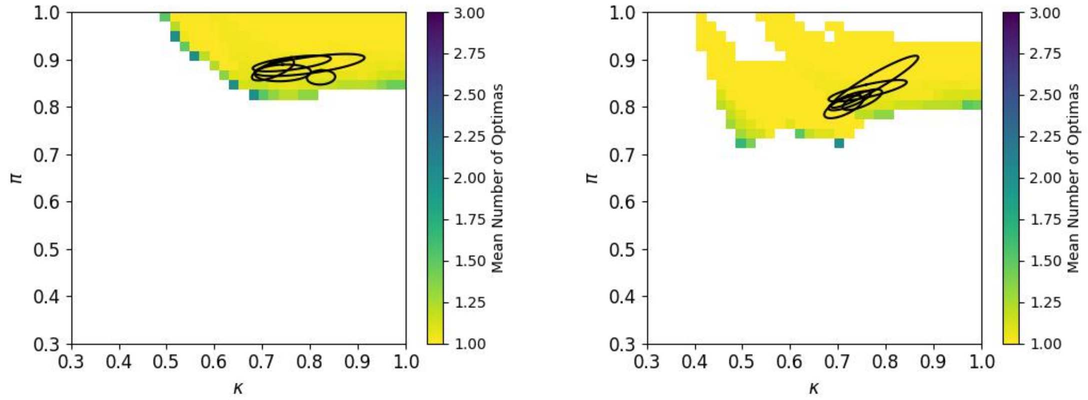
Figure 3 shows substantial overlap between the 95% confidence ellipsoid and the phase transition for a representative treatment; this holds across all four treatments (Figure A2 in Appendix C). As such, the data provide convincing evidence that market forces push prices so that the budget problem instance is in the phase transition, making it most difficult for consumers to determine whether they have reached the optimum. Prediction 7 is supported.

Prediction 9. We turn to the second Computational Requirement, which is that trade prices ensure that the budget problem instances have a unique optimum.

For the treatments where WE exists, Figure 4 superimposes the 95% confidence ellipsoids of the locations of the estimated long-run expectation of κs and πs on the color-coded map of mean number of optima (witnesses) of budget problem instances, constructed as in Figure 1c. We observe that the ellipsoids fully overlap with the yellow region, where 95% or more of the budget problem instances feature unique optima. The same obtains for the treatments where WE does not exist (see Figure A3 in Appendix C). Therefore, we have strong support for Prediction 9.

We take stock of the findings.

Interim Conclusion 4. *The data exhibit strong evidence in favor of Prediction 7,*



(a) **(b)**

(a) Treatment "Walras, C1." (b) Treatment "Walras, C2." $\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in shown treatment; p_i varying; Υ^* as in resulting optimum; 95% confidence ellipsoids based on (6). The color-coded map in Panel (a) is the same as the one in Figure 1.c. The two "No Walras" treatments are qualitatively similar and are shown in Figure A3 in Appendix C.

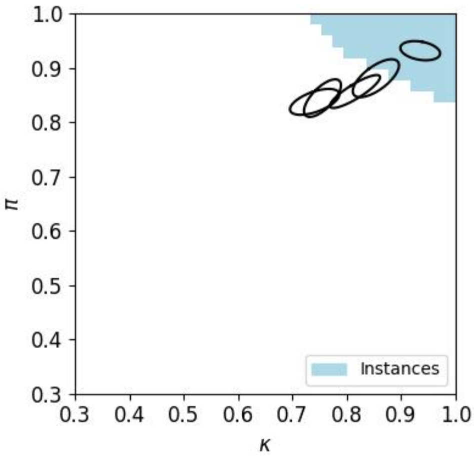
FIGURE 4 95% confidence ellipsoids of estimated asymptotic values for κ and π for each experimental session relative to mean number of witnesses (optima) in instances of budget problem, treatments where WE exists.

Prediction 8 and Prediction 9.

Computational versus Economic Requirements. We observe strong evidence in favor of economic equilibrium in the sense that in the vast majority of consumer-periods, consumers end up with one of the three equilibrium goods pairs in about equal fractions. We also find that utilities accruing to these pairs are significantly and sizably different. All this is economic support for CCE. On the computational side, we confirm that prices mostly force the budget problem instances to reside in the phase transition and in the region where most instances have a unique optimum, and hence, in the region of highest difficulty.

Yet prices are often *volatile*, to the point that they often push the budget problem instances into the region where one or more equilibrium goods pairs are unaffordable: between 1/2 and 2/3 of the trades occur at prices that make at least one equilibrium goods pair unaffordable. That is, trade takes place at prices that violate one of the key economic requirements, Prediction 4.

Figure 5 illustrates another way in which price volatility impacts the budget problem participants face. There, for a representative treatment, we plot the 95% confidence ellipsoid of κ and π on top of a map of the location of budget problems instances where all three equilibrium pairs are affordable, but not the triplet. The confidence ellipsoid tends to only partially overlap with the region where all goods pairs can be afforded, and sometimes not at all, suggesting that prices do not always tend to revert to levels that make all three pairs affordable; this pattern holds across the four treatments (Figure A4 in Appendix C).



Treatment "No Walras, C1." $\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in shown treatment; p_i varying; Υ^* as resulting optimum; 95% confidence ellipsoid based on (6). The remaining three treatments are qualitatively similar and are shown in Figure A4 in Appendix C.

FIGURE 5 95% confidence ellipsoid of estimated asymptotic values for κ and π, relative to the region where prices are such that all goods pairs can be afforded (blue area), representative treatment.

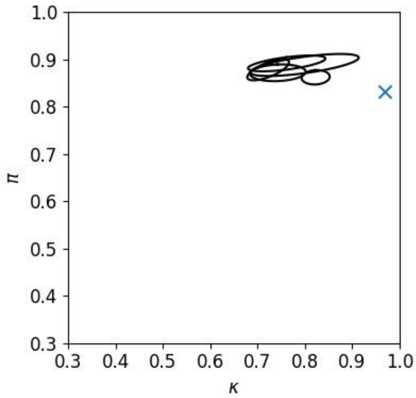
Price volatility causes consumers to be attentive: they need to track prices, solve their budget problem instance as soon as possible and act on this by submitting limit orders, or wait till prices revert back to those from budget problem instances they solved earlier and only then act. As a result, consumers apply cognitive effort not only to find optimal choices given trade prices, but also to pay attention to price changes, or to order flow, so that desired allocations can be implemented.

Interim Conclusion 5. *Price volatility makes it harder to attain maximum utility.*

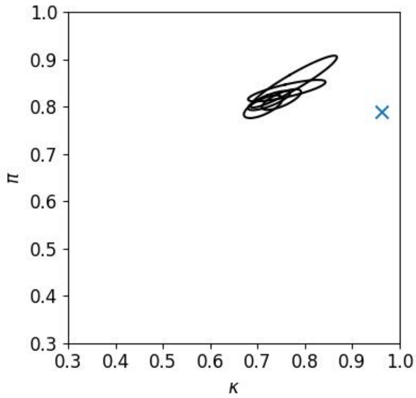
5.C Walrasian equilibrium

The final objective of our paper is to investigate the empirical support for WE in the Walras Treatment, where this equilibrium exists.

We have already rejected WE by verifying Prediction 6: we find that earnings associated with the three equilibrium goods pairs are statistically and sizably



(a)



(b)

(a) Treatment "Walras, C1." (b) Treatment "Walras, C2." $\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C+\sum_i(\phi_i - p_i))$; C, ϕ_i as in shown treatment; blue cross indicates the location of κ and π computed from p_i and Υ^* in WE; 95% confidence ellipsoids based on (6).

FIGURE 6 95% confidence ellipsoids of estimated asymptotic values for κ and π for treatments where WE exists, relative to location of WE

different, unlike in WE, where they necessarily have to be equal because everyone is assumed to fully optimize, and hence, whatever choices are made in equilibrium must earn the same.

At the same time, WE implies that the budget problem instance consumers face must have multiple (three) optima. However, we discover substantial overlap between the location in (κ, π) space of the budget problem instances that prices mean-revert to in the long run and the locations of budget problem instances that have mostly unique optima; see Figures 4a and 4b. That is, prices are such that the budget problem instances rarely if ever exhibit more than one optimum.

In fact, we can locate exactly where the WE resides in (κ, π) space. See Figure 6. The figure reveals that the WE is never inside any of the 95% confidence ellipsoids constructed from the experimental sessions. This is conclusive evidence against WE.

We can also directly test whether the trade prices are consistent with WE. Recall that WE prices are $(p_L, p_M, p_H) = (0.10, 1.30, 1.90)$ for payoff configuration C1, and $(p_L, p_M, p_H) = (0.10, 0.90, 1.70)$ for payoff configuration C2. Figure A5 in Appendix E displays the full time series of trade prices and compares them directly with the WE prices. For virtually all observations, the price levels of the lowest-value good, L, are five to ten times higher than the equilibrium level of 0.10.

We re-emphasize that prices are surprisingly high for another reason: the sellers of the goods pay zero marginal cost for the goods. In past experiments, this has consistently caused low trade prices, even prices below WE (Smith and Williams, 1982; Rasooly, 2022). In contrast, prices in CCE *have to be* high, so that the budget problem instances become sufficiently complicated. This suggests that computational complexity more than offsets the (opposite) force caused by the large rents that accrue to the supply side.

Interim Conclusion 6. *When it exists, WE is overwhelmingly rejected.*

6 Overall conclusions

When goods are indivisible, we find that markets set prices such that consumers' budget problem instance becomes computationally most complex. This complexity plays a central role in enabling market equilibration. When the budget problem instance consumers face is hard, agents with homogeneous preferences select different bundles because discovering the optimal allocation requires non-trivial cognitive effort or capability. Thus demand becomes heterogeneous, and this allows market clearance. Those who manage to identify better bundles are compensated with higher utility. The resulting Complexity Compensating Equilibrium (CCE) offers a framework in which equilibrium may exist even when WE does not.

In our experiment, computational complexity required prices to be markedly different from those predicted by WE (provided the latter existed). Prices located the budget problem instance in the intersection of the region where the instances live that tend to require most effort regardless of solution algorithm, i.e., in the "phase transition," and of the region where most instances had a unique optimum. Prices were high, in spite of the downward pressure from sellers who had zero marginal cost for supplying the goods. The data further supported CCE in that the various equilibrium choices implied significantly different utility levels. When WE existed, instead, all equilibrium bundles had to earn the same, and the budget problem instance had to have multiple optima.

We did discover an unexpected but powerful channel through which markets further raised the cognitive costs of reaching optimum utility levels, namely, price volatility. Among others, prices were sufficiently volatile so that at times the best goods bundle was not affordable, forcing those who expected to be able to buy this bundle to be more attentive and wait until prices reverted to levels where they could afford it, to quickly re-compute the optimum, or to settle for second-best.

In future work, we plan to increase the number of goods traded so that the budget problem instances become even more difficult.[35] We conjecture that the features of CCE will obtain without markets having to become volatile.

Future work should also aim at understanding the role of computational complexity (and other features that require higher cognitive effort or capability) in the presence of preference (payoff) heterogeneity. In this paper, because of homogeneity, the market can only use prices to ensure equilibration. With heterogeneity, the market can also exploit payoff diversity. We would still propose that the market chooses prices that put the budget problem instances in the phase transition. Because of payoff heterogeneity, however, the location of the phase transition changes. We conjecture that the phase transition will become more like that of the traditional KP. There, the phase transition consists of normalized profits (π) that are close to, and generally slightly above, normalized capacities (κ) (Yadav et al., 2020). This would then provide generic CCE price predictions that can be tested even in the field.

Past research in economics involving computational complexity has focused on the difficulty of computing equilibria; see, e.g., Rust (1996), or Judd (2001). Our analysis shifts the focus away from computational complexity at the market level and towards that of the agents trading in the markets. In an influential article, Hayek (1945) criticized the lack of appreciation among economists for the complexity individual agents face when solving budget (or production) problems even if they have to take market prices as given. However, Hayek's hypothesis was that market equilibrium would make individual agents' problems easier. Our results instead suggest the opposite conclusion: in the face of homogeneous preferences, computational complexity at the agent level is needed for markets to equilibrate.

The broader implication of our findings is that cognitive constraints – often viewed as biases or limitations – may, in fact, be integral to the equilibration of decentralized markets. When indivisibilities make equilibration difficult or impossible because of preference homogeneity, the computational complexity of the budget problem – an NP-hard problem because of the indivisibilities – can generate sufficient demand heterogeneity for markets to equilibrate.

[35] We remind the reader however that a budget problem with 3 goods is not necessarily simple for humans. It can readily be made as difficult as instances with eight goods. See Murawski and Bossaerts (2016).

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Supplemental Appendices

Appendix A Calculations for Walrasian equilibrium existence

Here we provide a general formula for existence of Walrasian equilibrium in an economy with three goods and two agents as described in Section 2. Denote with ϕ_L, ϕ_M, ϕ_H the payoffs of the three goods and p_L, p_M, p_H the corresponding prices. Denote the budget of each agent with C. By assumption, $\phi_L \leq \phi_M \leq \phi_H$. Assume unit demand and linear utility so that $U(x_1, x_2, x_3) = x_1(\phi_1 - p_1) + x_2(\phi_2 - p_2) + x_3(\phi_3 - p_3) + C$, where $x_i \in \{0, 1\}$.

All prices need to be positive and smaller than or equal to the payoff, otherwise no one would buy the goods. This gives us the following inequalities:

$$0 \leq p_L \leq \phi_L$$
$$0 \leq p_M \leq \phi_M$$
$$0 \leq p_H \leq \phi_H$$

All equilibrium bundles of goods need to lead to same utility. For markets to clear, the bundles need to be $\{L, M\}$, $\{L, H\}$ and $\{M, H\}$. Indifference between the equilibrium bundles would require:

$$U(\{L, M\}) = U(\{M, H\}) \Rightarrow p_H = (\phi_H - \phi_L) + p_L$$
$$U(\{M, H\}) = U(\{H, L\}) \Rightarrow p_M = (\phi_M - \phi_L) + p_L$$
$$U(\{H, L\}) = U(\{L, M\}) \Rightarrow p_H = (\phi_H - \phi_M) + p_M$$

Combining the equalities and inequalities, we get:

$$0 \leq p_L \leq \phi_L$$
$$\phi_M - \phi_L \leq p_M \leq \phi_M$$
$$\phi_H - \phi_L \leq p_H \leq \phi_H$$

The inequalities above provide a range of prices within which markets can clear. We additionally need to ensure that prices are such that all pairs of goods are affordable. To ensure affordability, we need:

$$p_L + p_M \leq C \Rightarrow p_L + (\phi_M - \phi_L) + p_L \leq C \Rightarrow p_L \leq \frac{C - (\phi_M - \phi_L)}{2}$$
$$p_M + p_H \leq C \Rightarrow p_M + (\phi_H - \phi_M) + p_M \leq C \Rightarrow p_M \leq \frac{C - (\phi_H - \phi_M)}{2}$$
$$p_H + p_L \leq C \Rightarrow p_H - (\phi_H - \phi_L) + p_H \leq C \Rightarrow p_H \leq \frac{C + (\phi_H - \phi_L)}{2}$$

Combining the last two sets of inequalities, we get the final restrictions on prices.

$$0 \leq p_L \leq \min\left\{\phi_L, \frac{C - (\phi_M - \phi_L)}{2}\right\}$$
$$\phi_M - \phi_L \leq p_M \leq \min\left\{\phi_M, \frac{C - (\phi_H - \phi_M)}{2}\right\}$$
$$\phi_H - \phi_L \leq p_H \leq \min\left\{\phi_H, \frac{C + (\phi_H - \phi_L)}{2}\right\}$$

The inequalities are satisfied if

$$C \geq (\phi_M - \phi_L)$$
$$C \geq (\phi_M - \phi_L) + (\phi_H - \phi_L)$$
$$C \geq \phi_H - \phi_L$$

Given that we assume $\phi_H \geq \phi_M \geq \phi_L$, a Walrasian equilibrium exists if

$$C \geq (\phi_M - \phi_L) + (\phi_H - \phi_L)$$

In the motivating example of Section 2, payoffs are $(\phi_L, \phi_M, \phi_H) = (1200, 2400, 3000)$ and the minimum budget for equilibrium existence is $C = 3000$. For this budget, the price inequalities collapse to a unique price vector of $(p_L, p_M, p_H) = (0, 1200, 1800)$. For any budget $C > 3000$, a multiplicity of equilibria exists of the form $(p_L, p_M, p_H) = (p, 1200 + p, 1800 + p)$, where p can be any allowable number satisfying the inequality for good L.

In our experimental sessions, we used two payoff configurations. Configuration C1 was identical to the one analyzed above, with quantities divided by 1,000. So, payoffs were $(\phi_L, \phi_M, \phi_H) = (1.20, 2.40, 3.00)$. There, the minimum budget for equilibrium existence is $C = 3.00$ and in that case equilibrium prices are $(p_L, p_M, p_H) = (0.00, 1.20, 1.80)$. For configuration C2, the payoffs were $(\phi_L, \phi_M, \phi_H) =$

$(1.60, 2.40, 3.20)$. The minimum budget for equilibrium existence is $C = 2.40$ and in that case equilibrium prices are $(p_L, p_M, p_H) = (0.00, 0.80, 1.60)$.

Appendix B NP hardness of the budget problem

We reduce the Knapsack Problem to the Budget Problem to prove *NP-hardness*. Reduction of a known *NP-complete* problem, such as the 0-1 Knapsack Problem, when completed in *polynomial time*, puts a lower bound on the complexity class of the problem one reduces it to (Cook, 1971; Levin, 1973).

The Budget Problem is known and important to economics and finance, with a wealth of knowledge about the real-world applications of the problem. Study of the computational complexity of the Budget Problem and the implications of this complexity is a new and emerging field.

This reduction is in many ways trivial. We would not claim that the Budget Problem is a new hard-problem, but to our knowledge there is no existing direct proof, only assumptions, that the (decision version of the) Budget Problem is *NP-complete* and this reduction serves to remedy this shortcoming.

B.A 0-1 Knapsack problem

The 0-1 Knapsack Problem (KP) was proven to be an *NP-complete* problem in Karp (1972). An instance of the 0–1 Knapsack *decision* problem consists of a set of items $i \in \{1, \ldots, I\}$, where each item has a value $v_i \in \mathbb{Z}_{\geq 0}$ and a weight $w_i \in \mathbb{Z}_{\geq 0}$, together with a capacity $K \in \mathbb{Z}_{\geq 0}$ and a target $T_K \in \mathbb{Z}_{\geq 0}$. The decision variables $x_i \in \{0, 1\}$ indicate whether item i is selected. All numerical inputs are encoded in binary. The problem asks whether there exists a selection $x \in \{0, 1\}^I$ satisfying the following Equation A1.

$$\sum_{i=1}^{I} x_i v_i \geq T_K, \tag{A1}$$

$$\text{subject to:} \sum_{i=1}^{I} x_i w_i \leq K,$$

where T_K is the target value to be reached.

B.B Budget problem

Consider an economy with I indivisible goods, each fully characterized by a *price* p_i and a *payoff* ϕ_i. Let C be the available budget (i.e., cash), T_B be the target value, and let $x_i \in \{0, 1\}$ denote if the good i is purchased or not. The Budget Problem is defined as the following:

$$\sum_{i=1}^{I} x_i \phi_i + \left(C - \sum_{i=1}^{I} x_i p_i \right) \geq T_B, \tag{A2}$$

$$\text{subject to:} \sum_{i=1}^{I} x_i p_i \leq C.$$

Note that, unlike the consumer problem of (Gilboa et al., 2021), we allow the decision-maker to gain utility from leftover budget (cash), which is perfectly divisible.

An instance of the Budget decision problem consists of a set of items $i \in \{1, \ldots, I\}$, where each item has a payoff $\phi_i \in \mathbb{Z}_{\geq 0}$ and a price $p_i \in \mathbb{Z}_{\geq 0}$, together with a budget $C \in \mathbb{Z}_{\geq 0}$ and a target $T_B \in \mathbb{Z}_{\geq 0}$.[36] The decision variables $x_i \in \{0, 1\}$ indicate whether item i is purchased, and unspent budget $C - \sum_{i=1}^{I} x_i p_i$ contributes to utility at face value. All numerical inputs are encoded in binary. The problem asks whether there exists a selection $x \in \{0, 1\}^I$ satisfying Equation A2.

B.C The reduction

We prove that the Budget Problem is NP-hard by giving a polynomial-time reduction from the 0-1 Knapsack Problem. In particular, given an arbitrary Knapsack instance, we construct a corresponding Budget Problem instance whose answer is "yes" if and only if the original Knapsack instance has answer "yes."

As an initial attempt, given a Knapsack instance, one might construct a Budget Problem instance by setting $\phi_i = v_i$, $p_i = w_i$, $C = K$, and choosing $T_B = T_K + \left(K - \sum_{i=1}^{I} x_i w_i \right)$. This would yield the following condition:

[36] In the main text, we used the symbol Υ for target value of the Budget Problem.

$$\sum_{i=1}^{I} x_i v_i + \left(K - \sum_{i=1}^{I} x_i w_i \right) \geq T_K + \left(K - \sum_{i=1}^{I} x_i w_i \right), \tag{A3}$$

$$\text{subject to:} \sum_{i=1}^{I} x_i w_i \leq K.$$

The $(K - \sum_{i=1}^{I} x_i w_i)$ cancel out and this is the Knapsack formula from Equation A1. Intuitively, what this is saying is that if you decrease the target in the Knapsack Problem by the amount of excess cash for a given solution, that will be identical to the Budget Problem. That makes sense because the only difference between the problems is that you get closer to the respective target in the Budget Problem for each excess cent you have, whereas in the knapsack this excess cash is discarded.

The issue is, however, that calculating $\sum_{i=1}^{I} x_i w_i$ (the amount of excess cash for the current solution) depends on the current solution, and while we could use the optimal solution to fix it, that has no known efficient (polynomial time) algorithm. So we cannot use it to set T_B to maintain the functional form of the Budget Problem.

Instead, what we will do is leave K, which is constant and known, as part of the substitution for T_B, but then add $\sum_{i=1}^{I} x_i w_i$ to both sides of the objective function.

$$\sum_{i=1}^{I} x_i v_i + \left(K - \sum_{i=1}^{I} x_i w_i \right) + \sum_{i=1}^{I} x_i w_i \geq T_K + \left(K - \sum_{i=1}^{I} x_i w_i \right) + \sum_{i=1}^{I} x_i w_i, \tag{A4}$$

$$\text{subject to:} \sum_{i=1}^{I} x_i w_i \leq K.$$

On the right hand side, this will cancel out, leaving us a known (and efficiently computable) expression. On the left hand side, we do not use it to cancel out, as this would change the function from the Budget Problem back to the Knapsack Problem, but instead merge it with the value of the items.

$$\sum_{i=1}^{I} x_i(v_i + w_i) + \left(K - \sum_{i=1}^{I} x_i w_i\right) \geq T_K + K, \tag{A5}$$

$$\text{subject to:} \sum_{i=1}^{I} x_i w_i \leq K.$$

We now have the same functional form in Equation A5 (the substituted Knapsack Problem) as Equation A2 (the Budget Problem), but using the parameters from the Knapsack Problem.

We summarize the necessary substitutions in the Table A1.

TABLE A1 Substitutions required to reduce the Knapsack Problem to the Budget Problem.

Budget Problem	**Knapsack Problem**
ϕ_i	$v_i + w_i$
p_i	w_i
C	K
T_B	$T_K + K$

Importantly, the construction can be carried out in time polynomial in the size of the input. In particular, it performs a constant number of arithmetic operations for each item, together with a constant number of operations on the capacity and target value, and is therefore linear in the encoded input size. This establishes that the mapping is a valid polynomial-time reduction, as required for proving NP-hardness.

Note that the transformation in Table A1 preserves the assumption on the Budget Problem of non-negative parameters; since $v_i, w_i, K, T_K \in \mathbb{Z}_{\geq 0}$, the constructed parameters $\phi_i = v_i + w_i$, $p_i = w_i$, $C = K$, and $T_B = T_K + K$ are also nonnegative integers, so the output is a valid instance of the Budget Problem.[37]

[37] There is a more intuitive transformation in the reverse direction: set each Knapsack value to the item's face value minus its price, $v_i = \phi_i - p_i$, and the Knapsack target to the target minus the budget, $T_K = T_B - C$. While answer-preserving, this does not ensure that the constructed values and target are nonnegative, and so its raw output does not in general meet the standard definition of the knapsack problem; a linear-time repair exists (dominated items with $\phi_i < p_i$ can be discarded, and instances with $T_B \leq C$ are trivially yes-instances), but note that this direction in any case only bounds the Budget Problem's complexity from above.

Using the above transformation the correctness is biconditional as required: any instance of the Knapsack Problem which is solvable (the answer is "yes"), will reduce into a Budget Problem which is also solvable. Any non-solvable Knapsack Problem instance (the answer is "no"), will reduce into a Budget Problem which is also non-solvable. Here is the proof.

Theorem 1. *The decision version of the Budget Problem is NP-hard.*

Proof. Consider the reduction from the decision version of the 0-1 Knapsack Problem described above. Under this construction,

$$\sum_{i=1}^{I} x_i p_i \leq C \quad \iff \quad \sum_{i=1}^{I} x_i w_i \leq K,$$

so feasibility is preserved. Moreover,

$$\begin{aligned} \sum_{i=1}^{I} x_i \phi_i + \left(C - \sum_{i=1}^{I} x_i p_i \right) &= \sum_{i=1}^{I} x_i (v_i + w_i) + \left(K - \sum_{i=1}^{I} x_i w_i \right) \\ &= \sum_{i=1}^{I} x_i v_i + K. \end{aligned}$$

Since $T_B = T_K + K$, the Budget target is met if and only if the Knapsack target is met. Hence the constructed Budget instance is a yes-instance if and only if the original Knapsack instance is a yes-instance.

The construction runs in polynomial time. Therefore, 0–1 KNAPSACK $\leq_p$ BUDGET (formal notation for "Knapsack Problem instances can be reduced in polynomial time to Budget Problem instances), so the Budget Problem is NP-hard. □

In fact, the reduction preserves witnesses exactly: the same selection x works in both instances. General reductions need only preserve the yes/no answer, so this is a stronger property than required.

B.D Membership in NP

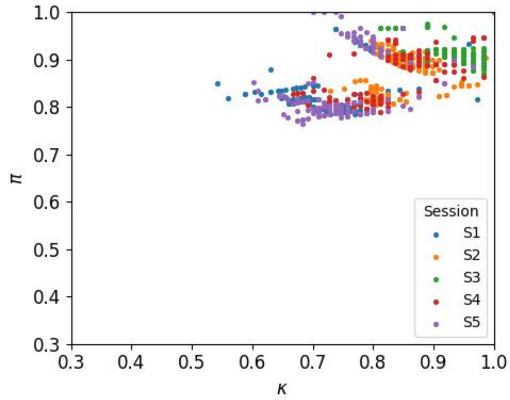
Now that we have proven *NP-hardness*, to prove it is *NP-complete*, we also need to show the Budget Problem is in *NP*.

This is trivial; a witness x can be verified in polynomial time by checking $\sum_{i=1}^{I} x_i p_i \leq C$ and evaluating $\sum_{i=1}^{I} x_i \phi_i + \left(C - \sum_{i=1}^{I} x_i p_i \right) \geq T_B$, each of which requires $O(I)$ arithmetic operations.

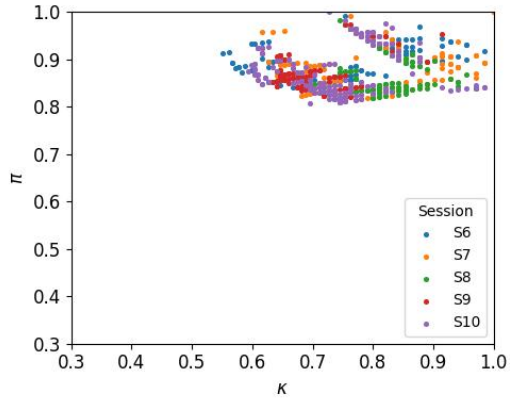
As such, the Budget Problem is in *NP*, is *NP-hard*, and therefore is *NP-complete*.

Appendix C Additional treatment-level figures

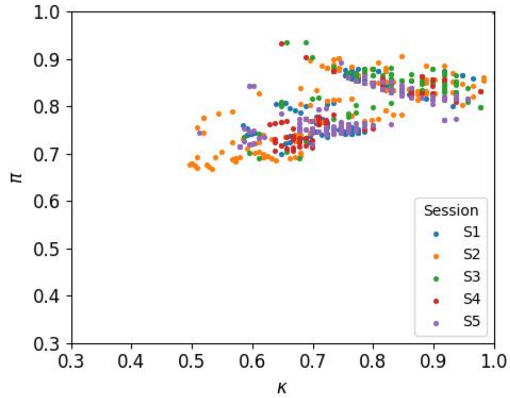
This appendix reports, for all four experimental treatments, the full versions of figures that are shown in Section 5 for a single representative treatment (or, in the case of Figure A3, for the two remaining "No Walras" treatments). In all cases, the omitted treatments are qualitatively similar to the one(s) reported in the main text.



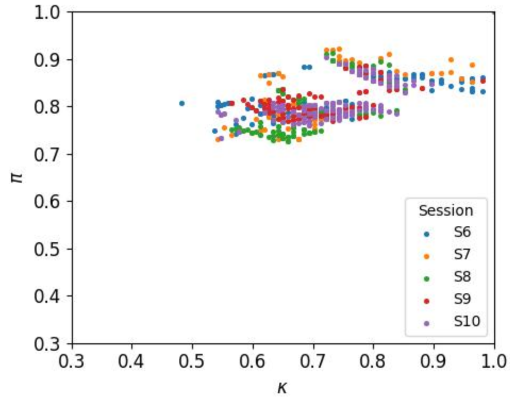
(a)



(b)



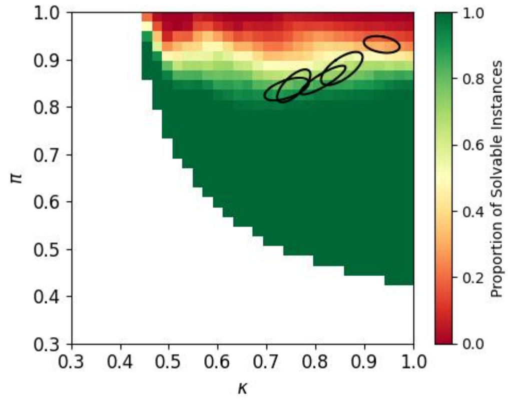
(c)



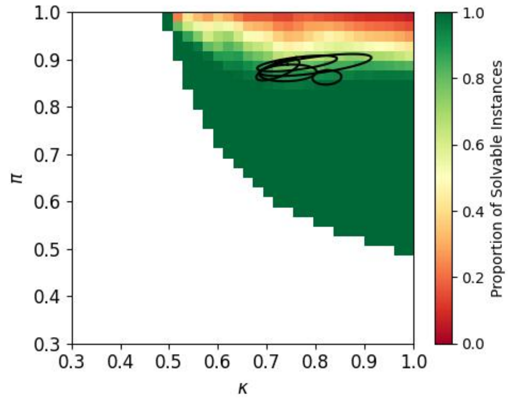
(d)

$\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i, p_i, and Υ^* as in indicated sessions. Each dot corresponds to a trade. (a) Treatment "No Walras, C1" (Sessions S1 to S5). (b) Treatment "Walras, C1" (Sessions S6 to S10). (c) Treatment "No Walras, C2" (Sessions S1 to S5). (d) Treatment "Walras, C2" (Sessions S6 to S10).

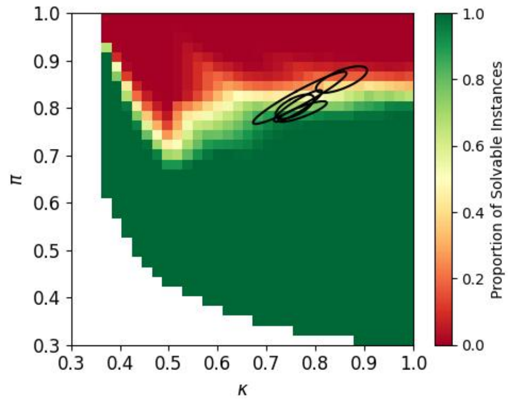
FIGURE A1 Recorded locations of the budget problem in $\kappa \times \pi$ space, all treatments. Compare with Figure 2.



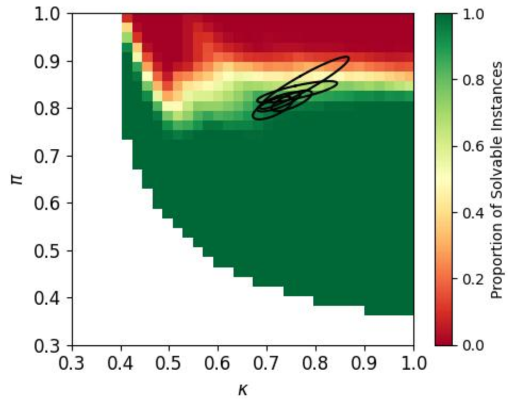
(a)



(b)



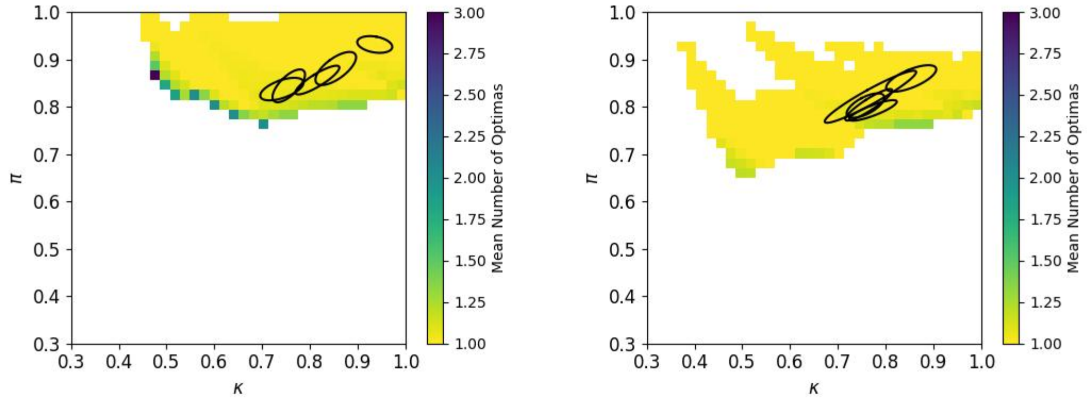
(c)



(d)

$\kappa = C/\sum_i p_i$; $\pi = \Upsilon/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in indicated Treatment; p_i, and Υ varying; 95% confidence ellipsoids based on (6), where p_i equals trade price and Υ is value at the optimum choice. (a) Treatment "No Walras, C1." (b) Treatment "Walras, C1." (c) Treatment "No Walras, C2." (d) Treatment "Walras, C2." Phase transitions are indicated in light green down to orange. The color-coded map in Panel (b) is the same as the one in Figure 1.a.

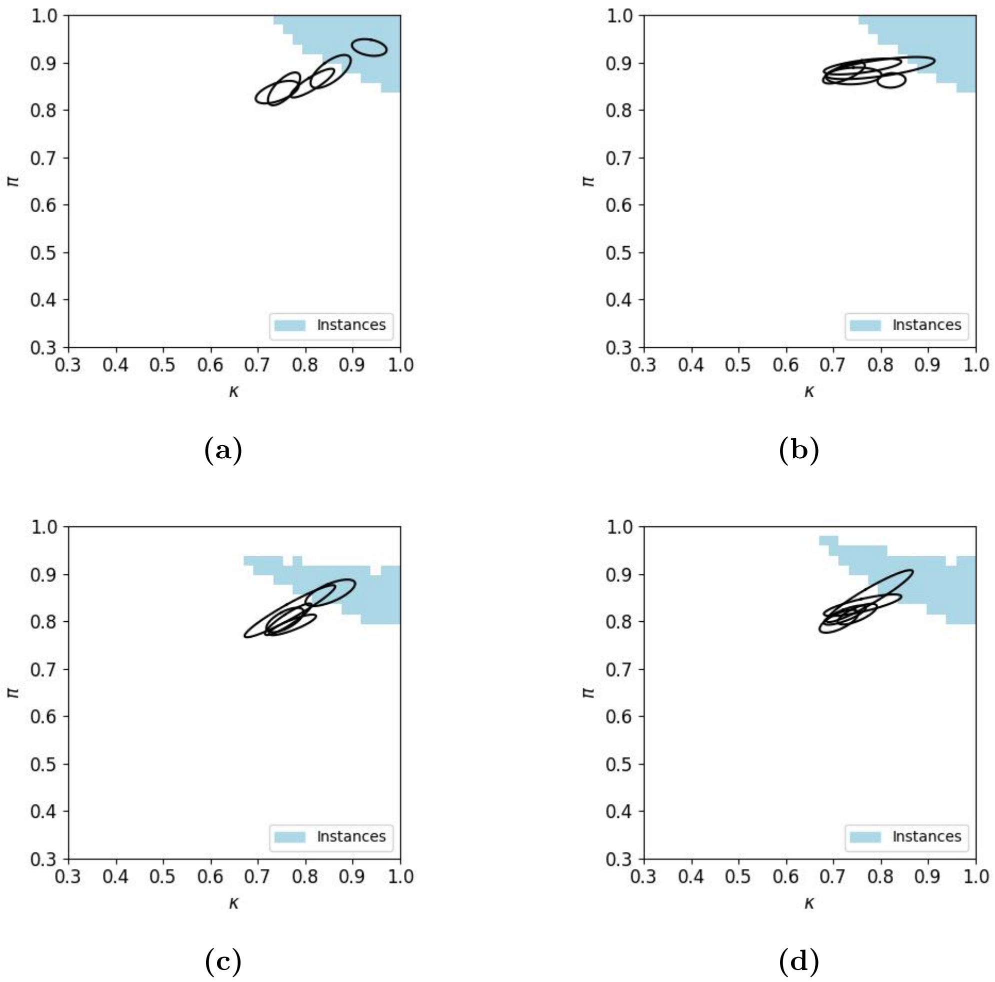
FIGURE A2 95% confidence ellipsoids of estimated asymptotic values for κ and π in each experimental session, arranged per treatment, relative to the phase transition map of the budget problem, all treatments. Compare with Figure 3.



(a) (b)

$\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in indicated Treatment; p_i varying; Υ^* as resulting optimum; 95% confidence ellipsoids based on (6). (a) Treatment "No Walras, C1." (b) Treatment "No Walras, C2." Confidence ellipsoids overlay the mean number of witnesses. The corresponding "Walras" treatments are shown in Figure 4.

FIGURE A3 95% confidence ellipsoids of estimated asymptotic values for κ and π for each experimental session, relative to mean number of witnesses (optima) in instances of budget problem, "No Walras" treatments. Compare with Figure 4.



$\kappa = C/\sum_i p_i$; $\pi = \Upsilon^*/(C + \sum_i(\phi_i - p_i))$; C, ϕ_i as in indicated Treatment; p_i varying; Υ^* as resulting optimum; 95% confidence ellipsoids based on (6). (a) Treatment "No Walras, C1." (b) Treatment "Walras, C1." (c) Treatment "No Walras, C2." (d) Treatment "Walras, C2." The blue region in Panel (b) is the same as the one in Figure 1.d.

FIGURE A4 95% confidence ellipsoids of estimated asymptotic values for κ and π for each experimental session, arranged per treatment, relative to the region where prices are such that all goods pairs can be afforded (blue area), all treatments. Compare with Figure 5.

Appendix D More details on goods choices in the experiment

In this appendix, we provide a more detailed breakdown of choices in our experiment.

We first look at the seller side of the markets. At the beginning of each period, each seller is endowed with two units of each good, so six goods in total. Table A2 shows the distribution of sold and unsold units across treatments, using both seller-periods and the actual number of goods as the unit of observation. We see that the vast majority of sellers successfully sold all their goods in all periods (519/720 seller-periods), and generally almost all units were sold (3,882/4,320). Efficiency is calculated as the percentage of units sold as shown in the last three columns of the table.

TABLE A2 Distribution of sold and unsold units of goods across treatments

Budget Treatment	Payoff Treatment	Units of goods sold per seller-period 6	5	4	3	2	1	0	Total	Number of Units of goods Sold	Unsold	Total
Low	C1	143	13	10	6	2	2	-	176	987	69	1,056
	C2	120	28	16	6	1	5	-	176	949	107	1,056
High	C1	129	26	11	9	4	4	1	184	987	117	1,104
	C2	127	23	11	9	3	5	6	184	959	145	1,104
Aggregate		519	90	48	30	10	16	7	720	3,882	438	4,320

We move on to the consumer side of the markets. Table A3a presents the full distribution of final goods holdings across all consumer-periods. We observe that the majority (68.6%) of consumers end up with a pair of goods. In rare occasions, consumers manage to buy all three goods (3.4%). This leaves less supply for other consumers, resulting in 19.6% of consumers buying a single good and 3.9% of consumers not buying any good at all.

Remarkably, in rare cases (4.5%) consumers end up holding more than one unit of a good. This can sometimes be by mistake, but it is also possible that some consumers are buying excess units of a good when the price is low, speculating that they will be able to sell later when the price is high (this was allowed). Since they did not successfully resell the excess units, those consumers will not be paid a liquidating dividend for their additional units of holdings. Thus, such excess demand decreases welfare as it prevents other consumers from increasing their utility.

For equilibrium to obtain, consumers need to end up with pairs of goods in equal proportions. The upper panel shows a roughly similar proportion of consumers purchasing each goods pair, suggesting that equilibrium allocations are reached. There is some mismatch between treatments with more consumers purchasing goods pair $\{M, L\}$ in the high-budget treatment and more consumers purchasing pair $\{H, M\}$ in the low-budget treatment. Table A3b shows the percentage of consumer-periods purchasing each good, aggregated over all possible

TABLE A3 Distribution of goods purchased per consumer-period

Category	Goods	No Walras-C1		No Walras-C2		Walras-C1		Walras-C2		Aggregate	
		Count	%	Count	%	Count	%	Count	%	Count	%
Triplet	{H,M,L}	32	6.1%	19	3.6%	10	1.8%	13	2.4%	74	3.4%
Pairs	{M,L}	94	17.8%	115	21.8%	111	20.1%	143	25.9%	463	21.4%
	{H,L}	127	24.1%	130	24.6%	147	26.6%	119	21.6%	523	24.2%
	{H,M}	143	27.1%	130	24.6%	124	22.5%	98	17.8%	495	22.9%
	Subtotal	364	68.9%	375	71.0%	382	69.2%	360	65.2%	1481	68.6%
Singles	{L}	21	4.0%	19	3.6%	20	3.6%	13	2.4%	73	3.4%
	{M}	32	6.1%	30	5.7%	38	6.9%	27	4.9%	127	5.9%
	{H}	26	4.9%	51	9.7%	49	8.9%	98	17.8%	224	10.4%
	Subtotal	79	15.0%	100	18.9%	107	19.4%	138	25.0%	424	19.6%
Excess	{H,H}	1	0.2%	0	0.0%	0	0.0%	1	0.2%	2	0.1%
	{H,M,M}	0	0.0%	1	0.2%	0	0.0%	0	0.0%	1	0.0%
	{H,L,L}	6	1.1%	3	0.6%	4	0.7%	0	0.0%	13	0.6%
	{H,L,L,L}	0	0.0%	0	0.0%	3	0.5%	0	0.0%	3	0.1%
	{M,M}	4	0.8%	1	0.2%	9	1.6%	7	1.3%	21	1.0%
	{M,M,L}	0	0.0%	2	0.4%	3	0.5%	5	0.9%	10	0.5%
	{M,M,L,L}	0	0.0%	1	0.2%	0	0.0%	0	0.0%	1	0.0%
	{M,L,L}	11	2.1%	4	0.8%	3	0.5%	9	1.6%	27	1.3%
	{M,L,L,L}	2	0.4%	0	0.0%	2	0.4%	0	0.0%	4	0.2%
	{M,L,L,L,L}	1	0.2%	0	0.0%	1	0.2%	0	0.0%	2	0.1%
	{L,L}	3	0.6%	3	0.6%	2	0.4%	2	0.4%	10	0.5%
	{L,L,L}	0	0.0%	0	0.0%	1	0.2%	0	0.0%	1	0.0%
	{L,L,L,L}	1	0.2%	0	0.0%	0	0.0%	0	0.0%	1	0.0%
	{L,L,L,L,L}	0	0.0%	0	0.0%	1	0.2%	0	0.0%	1	0.0%
	Subtotal	29	5.5%	15	2.8%	29	5.3%	24	4.3%	97	4.5%
None	∅	24	4.5%	19	3.6%	24	4.3%	17	3.1%	84	3.9%
Total		528	100%	528	100%	552	100%	552	100%	2,160	100%

(a) Consumer end-of-period holdings

Category	Goods	No Walras-C1		No Walras-C2		Walras-C1		Walras-C2		Aggregate	
		Count	%	Count	%	Count	%	Count	%	Count	%
Holds good	{H*}	335	35.2%	334	35.8%	337	35.6%	329	35.2%	1,335	35.4%
	{M*}	319	33.5%	303	32.5%	301	31.8%	302	32.3%	1,225	32.5%
	{L*}	298	31.3%	296	31.7%	308	32.6%	304	32.5%	1,206	32.0%
Total		952	100.0%	933	100.0%	946	100.0%	935	100.0%	3,766	100.0%

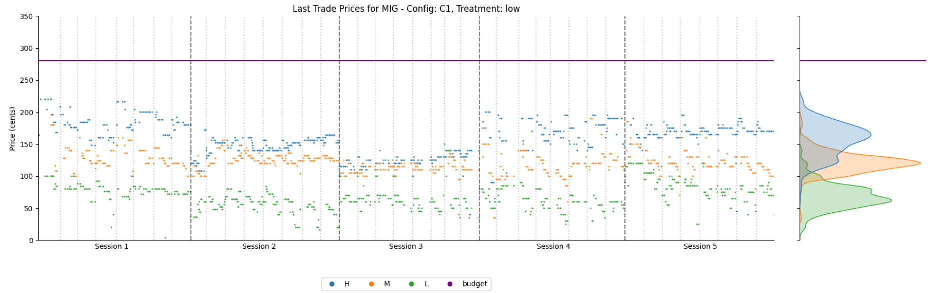
(b) Distribution of individual goods across consumers

bundles. For example in 335 out of 952 of consumer-periods in the first treatment (column 3), consumers purchased good H (as a single good or together with other goods). The proportions of consumers purchasing each individual good is statistically indistinguishable from equal proportions for all treatments.[38]

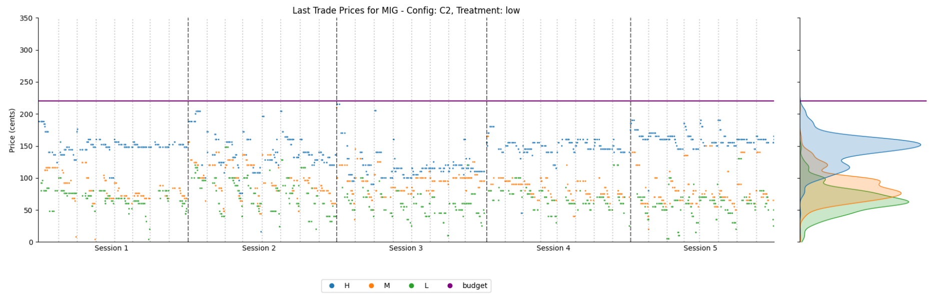
[38]Using a multinomial logistic model with random intercepts at the consumer and session level and estimating it separately per treatment, we get the following results. For No Walras-C1 treatment, $\mathcal{X}^2(1) = 2.170, p = 0.338, N = 952$. For No Walras-C2 treatment, $\mathcal{X}^2(1) = 2.630, p = 0.268, N = 933$. For Walras-C1 treatment, $\mathcal{X}^2(1) = 2.311, p = 0.315, N = 946$. For Walras-C2 treatment, $\mathcal{X}^2(1) = 1.452, p = 0.484, N = 935$. For aggregate data, $\mathcal{X}^2(1) = 7.728, p = 0.021, N = 3,766$.

Appendix E More details on goods prices in the experiment

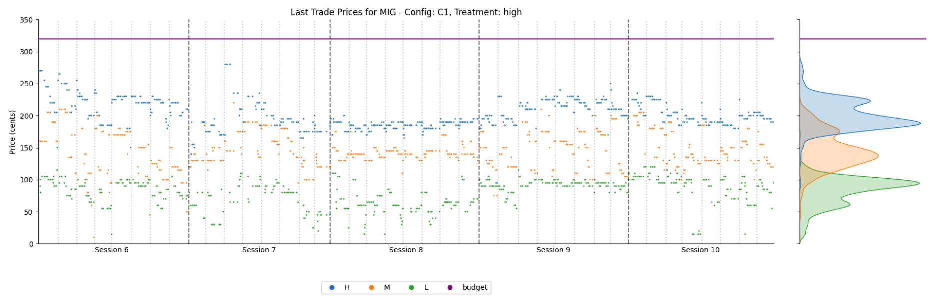
Here, we provide more details of the evolution of prices. We do so with two figures. Figure A5 plots the raw time series of the prices of the three goods for each treatment. We note the following observations: (i) prices are very volatile, increasing the complexity the consumers are facing, (ii) prices are higher when consumers have a high budget (i.e., when WE exists), (iii) prices of good L are 5-10 times higher than £0.10, suggesting large deviations from Walrasian equilibrium. Figure A6 plots the prices of pairs and the triplet of goods for all treatments. The reader can easily verify that consumers could rarely afford all goods together. This means that their budget constraint was binding and that they were solving a computationally hard decision problem.



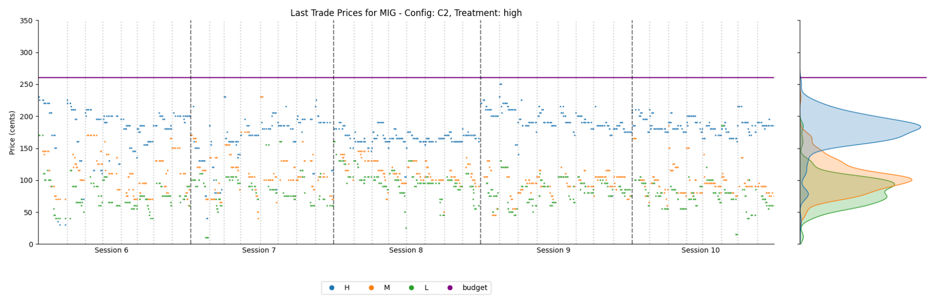
(a) **No Walras, Payoff Configuration C1**



(b) **No Walras, Payoff Configuration C2**



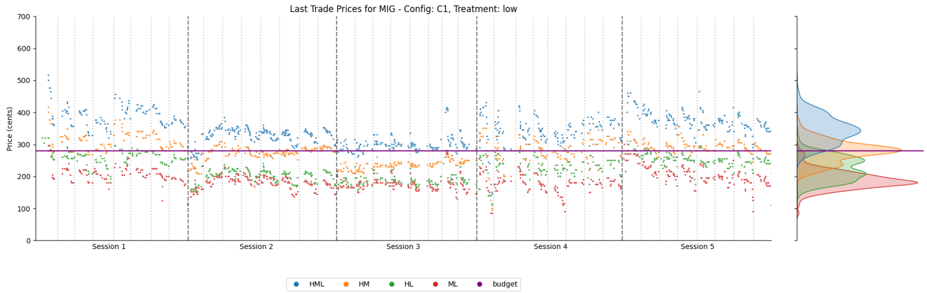
(c) **Walras, Payoff Configuration C1**



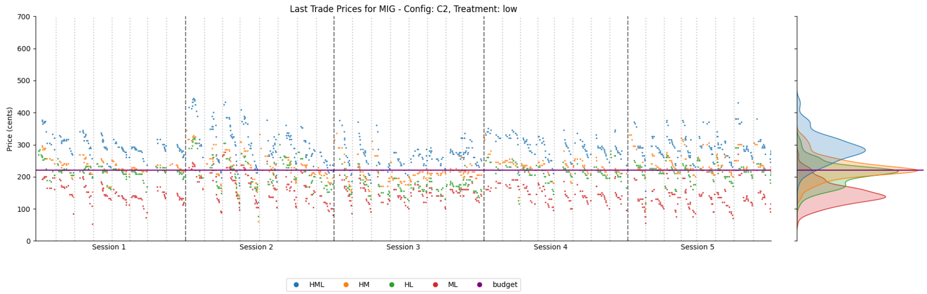
(d) **Walras, Payoff Configuration C2**

Notes: Dotted vertical lines separate experimental sessions. Purple horizontal line shows consumer budget level. Right panel of each figure shows the histogram of prices. A price is shown whenever there is a trade. Only one good traded; prices of other goods are obtained as last traded price.

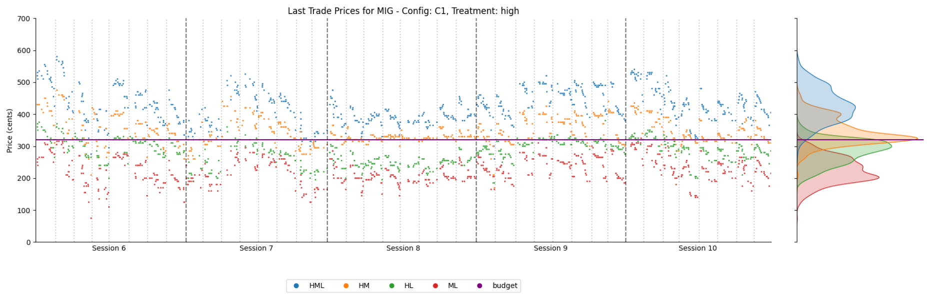
FIGURE A5 Time-series plot of good trade prices, by treatment



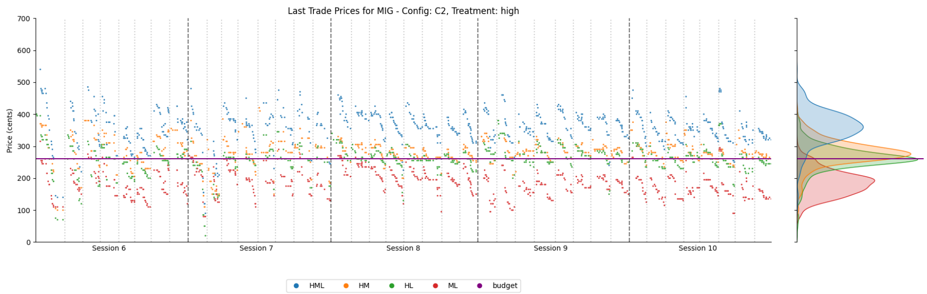
(a) No Walras, Payoff Configuration C1



(b) No Walras, Payoff Configuration C2



(c) Walras, Payoff Configuration C1



(d) Walras, Payoff Configuration C2

Notes: Dotted vertical lines separate experimental sessions. Purple horizontal line indicates consumer budget level. Right panel of each figure shows histogram of prices. A price is shown whenever there is a trade. Only one good traded; prices of other goods are obtained as last traded price.

FIGURE A6 Time-series plot of prices of various goods bundles, by treatment

Appendix F Experiment participant instructions

Notes: (i) Original instructions referred to goods as "assets"; (ii) Across sessions, the only change in the instructions was the Consumer First Good Values (payoffs) table.

Summary

You will trade three goods and cash in an anonymous online marketplace with other participants. You will have one of two goals. You may start with cash but no goods, in which case your goal is to acquire **one** unit in as many goods as you can, for as little cash as possible. You may start with goods but no cash, in which case your goal is to sell **all** your goods, for as much cash as possible.

This game will be played multiple times. A replication will be referred to as a "period." Periods are identical, except that you may start with different initial endowments and the value of goods may change. Your take-home pay will depend on your earnings from four (4) randomly drawn periods out of all the periods we run.

Trading Game

In the trading game, in each period you will be assigned the role of either a **Seller** or a **Consumer**. You can find out which you are by looking at your holdings. This is important, so please note it each period.

A **Seller** will start with a positive quantity of the goods, but no cash. A Seller gets **NO** value from holding goods, but they do for cash, so they want to sell their goods for as much cash as possible. Any goods not sold by a Seller at the end of the period will be discarded for no value.

A **Consumer** starts with cash, but no goods. goods provide significant value to the Consumers, so Consumers will likely want to use their cash to purchase goods. Importantly, Consumers only get value from the FIRST unit of each good they hold, plus any remaining cash they have left over. A Consumer holding two or more units of an good will not collect value for the subsequent units; only the first unit will count.

The three goods are called A, B, and C, and their **Consumer First Good Values** can be found below:

Consumer First Good Values	A	B	C
Periods 1-8 (and practice)	£1.20	£2.40	£3.00
Periods 9-16	£2.40	£3.20	£1.60

In the trading platform, currency will be expressed in experimental dollars, referred to with the symbol \$. This is just the software that we use. \$1.00 experimental dollar is equivalent to £1.00 British Pound.

Trading

Online Marketplace

In this experiment, you are asked to trade with other participants online through a platform called Flex-E-Markets. This will be opened and signed in on the computer in front of you. We will spend time demonstrating the platform soon, but for now we need to focus on the rules of trading. However, if you have any issues or are signed out during the experiment, raise your hand and an experimenter will sign you back in.

Trading Protocol

Trading takes place as follows. You submit limit orders: orders to buy a unit of an good at a chosen price (or lower), or to sell a unit at a chosen price (or higher). Transactions take place from the moment a buy order with a higher (or equal) price crosses a sell order with a lower (or equal) price or the other way around. Buy orders are coloured blue; sell offers are coloured red.

<u>All trade occurs at the price specified by the best standing order.</u> In other words, if a trade occurs, the price of the earlier best order determines the price. Orders at a better price execute first. Given a price, orders arriving earlier execute first. Orders remain valid until you cancel them, or the marketplace closes.

We will now demonstrate the trading platform. After, you will be given sufficient time to practice submitting and cancelling orders. We will then complete the instructions, before running four full practice periods of the Trading Game.

Numerical Examples

Below we provide a few numerical examples to illustrate how your period earnings will be calculated.

1. Imagine you are a **Consumer**, and you start with \$4.00 of cash. The Consumer First Good Value of good A is \$1.20, the Consumer First Good Value of good B is \$2.40, and the Consumer First Good Value of good C is \$3.00. During the trading period,

    (a) you bought one unit of good A for \$0.80 and one unit of good B for \$2.00. In this case, your final portfolio is worth \$1.20 (from the first unit of good A) + \$2.40 (from the first unit of good B) + \$4.00 (your starting cash) - \$0.80 (the cost to buy A) - \$2.00 (the cost to buy B) = \$4.80.

    (b) you bought two units of good A for \$0.80 each and one unit of good B for \$2.00. In this case, your final portfolio is worth \$1.20 (from the first unit of good A) + \$2.40 (from the first unit of good B) + \$4.00 (starting cash) - \$0.80 (buying A) - \$0.80 (buying A) - \$2.00 (buying B) = \$4.00. Note that the second unit of good A has no value to you, but you still paid \$0.80 for it (an \$0.80 loss).

    (c) you bought one unit of good C for \$3.25. In this case, your final portfolio is worth \$3.00 (from the first unit of good C), + \$4.00 (starting cash) - \$3.25 (buying C) = \$3.75 (less than you started with).

2. Imagine you are a **Seller**, and you start with 2 units of good A, 2 units of good B, and 2 units of good C. During the trading period,

    (a) You sold two units of good A for \$0.75 and \$0.85 respectively, two units of good B for \$2.10 and \$1.90 respectively, and two units of good C for \$1.20 and \$2.80 respectively. Your final payout would be \$0.75 + \$0.85 (from the two units of good A you sold) + \$2.10 + \$1.90 (from the two units of good B you sold) + \$1.20 + \$2.80 (from the two units of good C you sold) = \$9.60.

    (b) You sold two units of good A for \$0.80 each, two units of good B for \$2.00 each, and no units of good C. Your final payout would \$0.80 + \$0.80 (from the two units of good A you sold) + \$2.00 + \$2.00 (from the two units of good B you sold) = \$5.60. Note that the units of good C you did not sell has no value for you.

    (c) You sold no units at all. Your final payout would be \$0.00 as the goods have no value to you.

Take-Home Pay

There will be **four** (4) **practice periods** and **sixteen** (16) **experiment periods**. Periods will last **3 minutes** each.

At the end of the experiment, we will randomly draw four (4) periods from the experiment periods and pay your earnings for those periods. This means it is in your best interest to perform well in all periods, because you do not know which will be paid. Your earnings in those four periods will be converted to British Pounds at the exchange rate of 1:1 ($1 converts to £1). This performance pay will be added to a base pay of £15. We expect take-home pay to be anywhere between £20 and £40.

GOOD LUCK!

Appendix G Flex-E-Markets trading interface

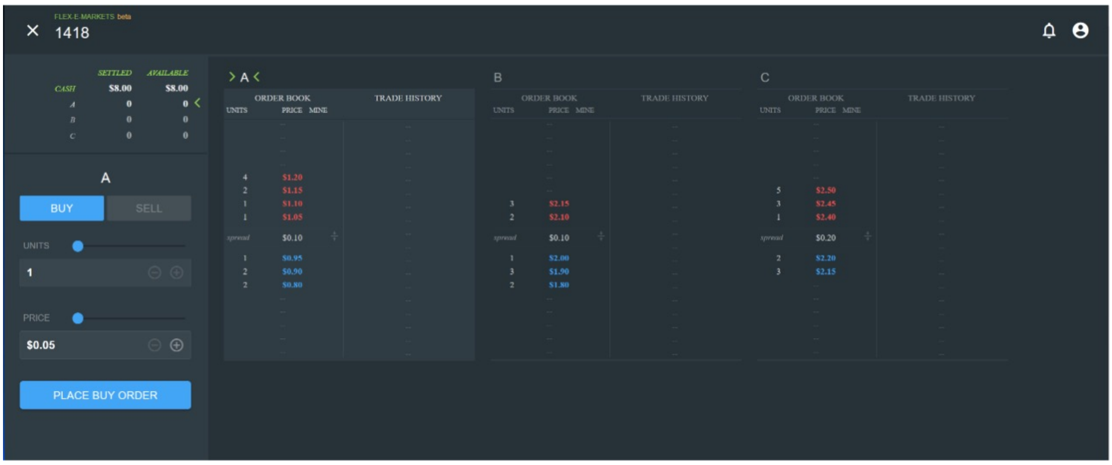


FIGURE A7 Flex-E-Markets Trading Interface

Appendix H Literature on equilibrium existence with indivisible goods

Ever since Henry (1970), the literature has attempted to uncover conditions under which equilibrium existence can be restored. Most existing approaches restore equilibrium by introducing heterogeneity, restricting preferences, or weakening market-clearing requirements. Note that the literature often studies the corresponding allocation problem, in which a central planner allocates goods to agents.

Under certain conditions, the second welfare theorem guarantees that an efficient allocation of the central-planner problem can be supported as a competitive equilibrium. While formally related, our approach focuses on the market representation of the problem because most goods markets are organized without a central planner, and because the computational complexity the market participants are facing plays a central role in our approach.

One strand of literature restores equilibrium by imposing structure on preferences. Without giving up homogeneity of preferences, the equilibrium existence is restored if the indivisible goods are gross substitutes (Kelso Jr and Crawford, 1982; Gul and Stacchetti, 1999), net substitutes (Danilov et al., 2001; Baldwin and Klemperer, 2019; Baldwin et al., 2023), Δ-substitutes, i.e., pairs of bundles that can be obtained by replacing up to Δ goods from each other (Nguyen and Vohra, 2024), or can be split into two groups with goods substitutable within groups but complementary between groups (Sun and Yang, 2006; Rostek and Yoder, 2020). Choi et al. (2018) provides a model that relies on preference heterogeneity. They study the case where consumers have partial knowledge of the payoffs and need to costly search to fully uncover them, and show that sufficient heterogeneity in prior valuations leads to a unique market equilibrium.

Another strand of literature restores equilibrium by relaxing the assumption of homogeneous budgets. Budish (2011) shows that if budgets are unequal but arbitrarily close and the number of traders is large, equilibrium can be restored in the limit. Babaioff et al. (2021) analyze budget perturbations that allow equilibrium to exist with a small number of traders.

A third approach relaxes the assumption of exact market clearing. This approach builds on the notion of social-approximate equilibria in which excess demand is bounded, and crucially relies on the assumption of large number of agents (Dierker, 1971; Broome, 1972; Svensson, 1984). This assumption is typically combined with relaxing preference homogeneity (Nguyen and Vohra, 2024), relaxing budget homogeneity (Budish, 2011), or by allowing agents to be allocated a lottery over goods (Budish et al., 2013; Gul et al., 2024; Gul and Pesendorfer, 2025).

A further approach in the literature has been to define alternative equilibrium concepts. In the context of markets for votes, Casella et al. (2012) propose an ex-ante type of competitive equilibrium: agents submit probabilistic demands and market clearing happens only in expectation. They impose a rationing rule to clear the market ex-post. Florig and Rivera (2017) propose a rationing equilibrium which relies on fiat money having a positive price and consumers having knowledge of the demand-supply imbalance. In the context of competitive lending, Asparouhova (2006) studies equilibrium existence by building on the model of Rothschild and Stiglitz (1976), who define an equilibrium as a set of loan contracts such that further contracts can no longer be introduced without causing losses. By expanding this to profitable addition of *bundles* of contracts, Asparouhova (2006) clarifies equilibrium existence and Pareto sub-optimality. She provides experimental evidence that the original Rothschild-Stiglitz equilibrium emerges only if it is Pareto optimal.

Our approach is closer in spirit to the latter strand of literature as we also define a new equilibrium concept. Our equilibrium is relevant in that it exists despite homogeneous preferences and budgets. We do not require any substitutability or complementarity of goods, and only assume additive preferences. The assumption of homogeneous preferences and budgets may seem extreme, but it is meant to be in opposition to the mainstream literature, which has appealed to sufficient preference or budget heterogeneity to assure equilibrium existence.

Our approach differs fundamentally from rational inattention and costly cognition models (Sims, 2006; Matějka and McKay, 2015), which assume agents optimally allocate attention subject to information-processing or cognitive costs. In those models, cognitive effort remains part of a standard optimization problem. In contrast, we study environments in which agents face NP-hard optimization problems where all primitives of the environment (prices, endowments, and payoffs) are fully observed, and focus on the difficulty of finding the optimum. Admittedly, agents might still fail to pay full attention to all inputs. However, past research has shown that human choice under computational complexity can be modeled successfully even disregarding potential inattention by solely appealing to mathematical features of the problem instance at hand (Andrabi et al., 2026). Moreover, even if some inputs are ignored, we doubt that inattention is determined in an optimal way. This is because identifying which inputs to not pay attention to is itself an NP-hard problem, the solution of which requires precise *a priori* knowledge of the types of budget instances anyone may encounter. One would then have to solve one (difficult-to-formulate) NP-hard problem in order to solve another (easily identifiable) NP-hard problem, a rather absurd situation.

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